

Article

Energy Demand Forecasting and Policy Development in Turkey

Ercan Köse ^{1,*}  and Sevil Kutlu Kaynar ²¹ Electrical-Electronics Engineering Department, Tarsus University, Tarsus, 33400 Mersin, Turkey² Energy Systems Engineering ABD, Graduate Education Institute, Tarsus University, Tarsus, 33400 Mersin, Turkey; 230913004@tarsus.edu.tr

* Correspondence: ekose@tarsus.edu.tr

Abstract

As Turkey's energy demand surges due to industrialization, population growth, and economic development, precise forecasting of electricity demand has become crucial for ensuring energy security and facilitating sustainable planning. This study undertakes an analysis of Turkey's current energy landscape and develops long-term electricity demand forecasts utilizing a diverse array of statistical and machine learning models, including linear regression, polynomial regression, and artificial neural networks (ANNs). By incorporating economic indicators, demographic trends, and historical consumption data, this research projects Turkey's electricity demand up to 2045. Among the various influencing factors, industrial production stands out as the most significant driver. The findings offer strategic insights into infrastructure investments, the integration of renewable energy, and policies aimed at enhancing efficiency. This research presents a data-driven, policy-oriented framework to assist decision-makers in reducing import dependence while steering Turkey towards a sustainable energy transition.

Keywords: electricity demand forecasting; regression models; artificial neural networks; renewable energy; energy policy; time series; Turkey



Academic Editors: Mirela Panait, Grzegorz Mentel and Xin Zhao

Received: 3 June 2025

Revised: 16 June 2025

Accepted: 19 June 2025

Published: 24 June 2025

Citation: Köse, E.; Kutlu Kaynar, S. Energy Demand Forecasting and Policy Development in Turkey. *Energies* **2025**, *18*, 3301. <https://doi.org/10.3390/en18133301>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

In recent decades, Turkey has experienced rapid economic growth, urbanization, and industrialization, resulting in a substantial increase in energy consumption. This trend, coupled with a high reliance on imported fossil fuels, has raised critical concerns regarding energy security, supply sustainability, and economic vulnerability. As energy demand continues to rise, particularly in the electricity sector, the development of robust forecasting models becomes crucial for infrastructure planning, investment strategies, and policy formulation.

This study offers a comprehensive analysis of Turkey's current energy structure and projects electricity demand up to 2045 using a variety of statistical and artificial intelligence-based models, including linear regression, polynomial regression, and artificial neural networks. By integrating economic, demographic, and technological variables, the research evaluates both short- and long-term dynamics of energy consumption and assesses the effectiveness of different forecasting methods.

This work not only focuses on demand modeling but also explores Turkey's energy policy initiatives, including the integration of renewable energy sources, strategies for enhancing energy efficiency, and sustainability efforts. It analyzes the transformation occurring in the industrial and transportation sectors, the growth of electric vehicles, and the national strategies aimed at reducing carbon emissions in alignment with global

environmental commitments, such as the European Green Deal. By providing both a data-driven forecast and a policy-oriented perspective, this study seeks to assist decision-makers in developing robust strategies that facilitate Turkey's energy transition toward a more secure, efficient, and sustainable future.

Investment requirements in the electricity sector for the upcoming period will be assessed considering unforeseen developments, including global and regional crises, epidemics, international political tensions, and factors related to energy supply security and the needs of the electricity grid. Appropriate additional measures will be implemented as necessary. According to the International Renewable Energy Agency's statistics on installed renewable energy capacity, Turkey holds the 11th position worldwide, with a total of 58,462 MW [1,2]. As illustrated in Figure 1, Turkey's energy scenario targets for 2020–2035 emphasize the planned trajectory of renewable energy development.

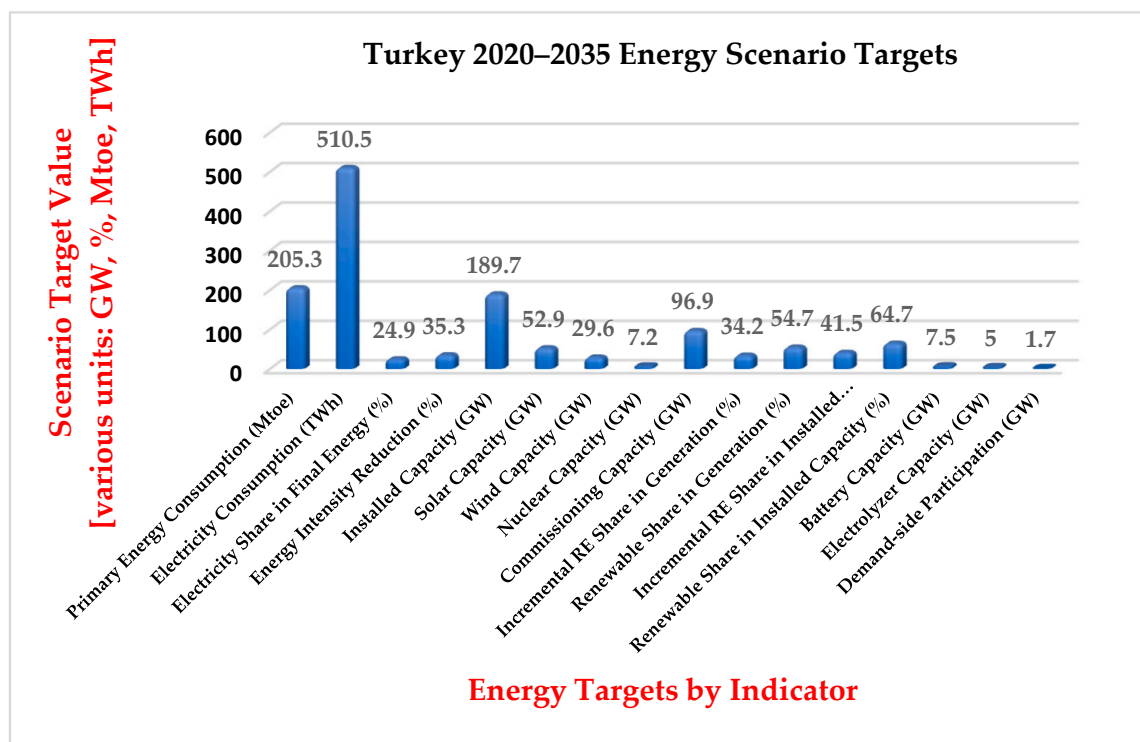


Figure 1. Turkey 2020–2035 period energy scenario targets [2].

Figure 2 illustrates the trend in domestic energy production in Turkey from 1990 to 2022. Since Turkey imports most of the energy it needs, it is affected by any fluctuations in the exchange rate, increases in energy prices, and geopolitical factors. These effects lead to fluctuations in energy costs. A large share of the production of the energy needed is provided using fossil energy resources. Natural gas, oil, and coal are the main sources preferred for energy production.

Turkey's energy landscape is significantly shaped by its dependence on imports, which exposes the country to fluctuations in the exchange rate, rising global energy prices, and geopolitical risks. These external pressures lead to instability in energy costs. Given that a significant portion of Turkey's energy supply still relies on fossil fuels, the need for strategic long-term planning is more relevant than ever. Addressing these structural vulnerabilities is crucial to ensuring a stable, secure, and sustainable energy future.

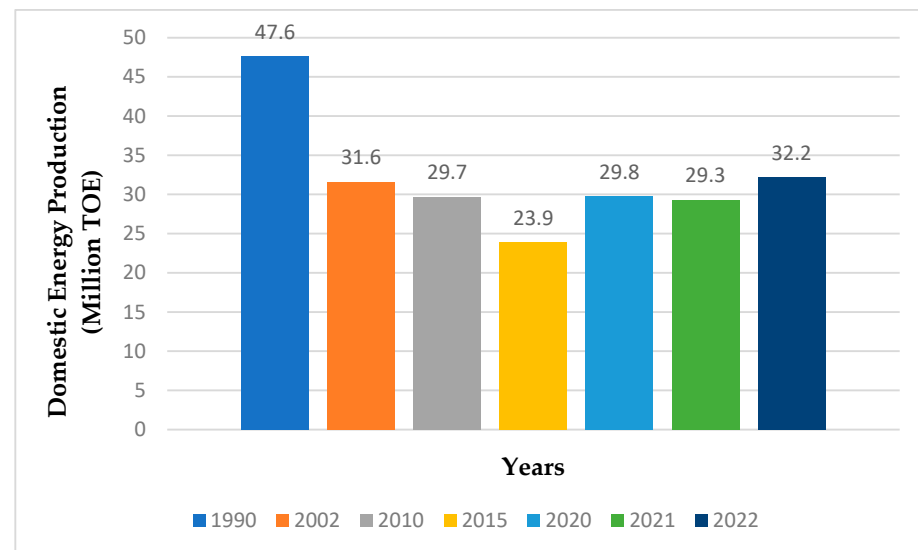


Figure 2. Domestic energy production in Turkey (1990–2022), million TOE [3].

Despite having limited domestic natural gas reserves, Turkey predominantly depends on imports, with natural gas-fired power plants playing a crucial role in electricity generation. As such, it can be concluded that Turkey is 99% reliant on imported natural gas. As of 2022, the technically recoverable natural gas reserves in Turkey are estimated to be around 544 billion m³ [4].

In 2022, our country produced 3.58 million tons of crude oil while importing 33.49 million tons. This data indicates that Turkey is 90% dependent on imported crude oil. As of 2022, Turkey's technically recoverable crude oil reserves are estimated to be around 70 million tons. Additionally, in that same year, a total of 191 crude oil wells were drilled, reaching a cumulative depth of 421,408 m [5].

In 2022, domestic natural gas production reached 408 million cubic meters, whereas imports totaled 54.6 billion cubic meters, reflecting a staggering 99% reliance on foreign sources. Similarly, crude oil production was recorded at 3.58 million tons, in stark contrast to imports of 33.49 million tons, illustrating a 90% dependency on imports. Despite possessing approximately 22.04 billion tons of coal reserves, comprising 20.53 billion tons of lignite and 1.51 billion tons of hard coal, primarily located in the Zonguldak region, the reliance on imported energy has increased significantly, rising from 52% in 1990 to 76% in 2015. According to the Turkish Hard Coal Enterprises Activity Report, domestic hard coal production continues to be limited despite significant reserves, which contributes to import dependency [6]. Recent growth in renewable energy sources, particularly solar, wind, and geothermal, has reduced dependency on imported energy, which fell to 70% in 2020 and 68% in 2022.

Most of Turkey's energy production is generated from thermal power plants, which primarily utilize fossil fuels such as coal, natural gas, and oil. Our country possesses domestic coal resources, and most of the thermal power plants operate using these supplies.

As of the end of October 2024, the distribution of our installed capacity by energy sources is as follows: 28.1% hydroelectric energy, 21.5% natural gas, 19.1% coal, 10.9% wind, 16.6% solar, 1.5% geothermal, and 2.4% other resources [7].

In preparing this report, the Ministry of Energy and Natural Resources considered various data points related to factors influencing electricity consumption, including the following:

- Economic growth rate;
- Population;

- Number of households;
- Contribution of the transport sector to electricity consumption;
- Internal consumption and grid losses;
- Efficiency.

Furthermore, the report incorporated energy data from the International Energy Agency (IEA) and sectoral Gross Domestic Product (GDP) figures from the World Bank for both Turkey and comparable countries. As illustrated in Figure 3 the projected annual average growth rates in electricity demand for the first decade, spanning 2019 to 2028, are anticipated to be 3.6%, 4.2%, and 4.8% for Scenarios 1, 2, and 3, respectively. Looking ahead to the subsequent 11 years from 2029 to 2039, these growth rates are estimated to be 2.4% for Scenario 1, 2.8% for Scenario 2, and 3.3% for Scenario 3 [8].

YEARS	Scenario 1	Scenario 2	Scenario 3	Scenario 1	Scenario 2	Scenario 3
	(GWh)	(GWh)	(GWh)	Change (%)	Change (%)	Change (%)
2019	313.832	315.182	316.503	4.6	5.0	5.5
2020	327.285	329.603	332.057	4.3	4.6	4.9
2021	340.511	344.407	348.662	4.0	4.5	5.0
2022	353.200	359.593	366.385	3.7	4.4	5.1
2023	366.767	375.821	385.177	3.8	4.5	5.1
2024	380.401	392.105	404.287	3.7	4.3	5.0
2025	392.610	406.939	422.303	3.2	3.8	4.5
2026	404.628	421.754	440.654	3.1	3.6	4.3
2027	416.619	436.632	458.917	3.0	3.5	4.1
2028	428.791	451.729	477.553	2.9	3.5	4.1
2029	441.026	466.820	496.555	2.9	3.3	4.0
2030	452.951	481.721	515.370	2.7	3.2	3.8
2031	464.567	496.702	533.989	2.6	3.1	3.6
2032	476.288	511.642	552.912	2.5	3.0	3.5
2033	487.827	526.353	571.626	2.4	2.9	3.4
2034	499.262	541.044	590.223	2.3	2.8	3.3
2035	510.826	555.682	608.452	2.3	2.7	3.1
2036	522.668	570.795	627.037	2.3	2.7	3.1
2037	534.008	585.308	644.636	2.2	2.5	2.9
2038	545.145	599.382	662.508	2.1	2.4	2.7
2039	556.324	613.386	679.865	2.1	2.3	2.6

Figure 3. Electricity demand forecast for Turkey (2020–2040) under three growth scenarios [8].

According to the key scenario data utilized in the population projections by the Turkish Statistical Institute, Turkey’s population is anticipated to grow by approximately 11.762% by 2044. Turkey’s population is anticipated to grow by approximately 11.762% by 2044, as shown in Figure 4. This population growth has led to a corresponding increase in energy demand. The trend of urbanization has particularly underscored the need for heating and electricity in highly urbanized regions. Examining the figures for per capita electricity consumption, Turkey’s gross electricity consumption per capita was approximately 1006 kWh in 1990, which rose significantly to 3931 kWh by 2021 [9,10].

The transport sector, which constitutes the third-largest share of Turkey’s final energy consumption, has experienced the most rapid growth in energy demand over the past

two decades, with an annual average growth rate of 4.5%. The ongoing global integration process, technological advancements, urban population concentration, and economic development have all contributed to an increased demand for high-quality, safe, and comfortable transportation services. As a result, the transport sector has become a dynamic and rapidly growing field, making efficient energy utilization essential. This modal distribution of transport energy consumption is detailed in Figure 5.

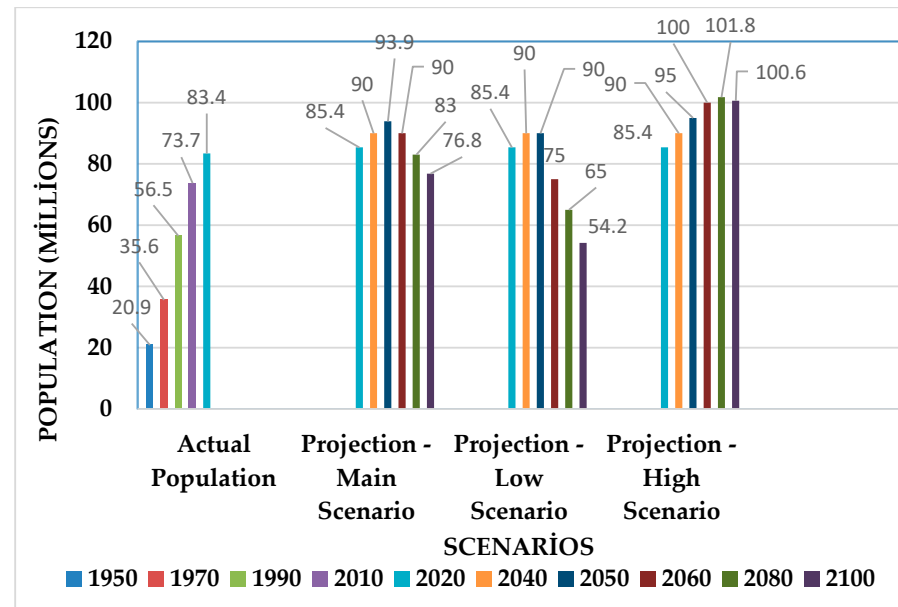


Figure 4. Population and population projections by year, 1950–2100 [9].

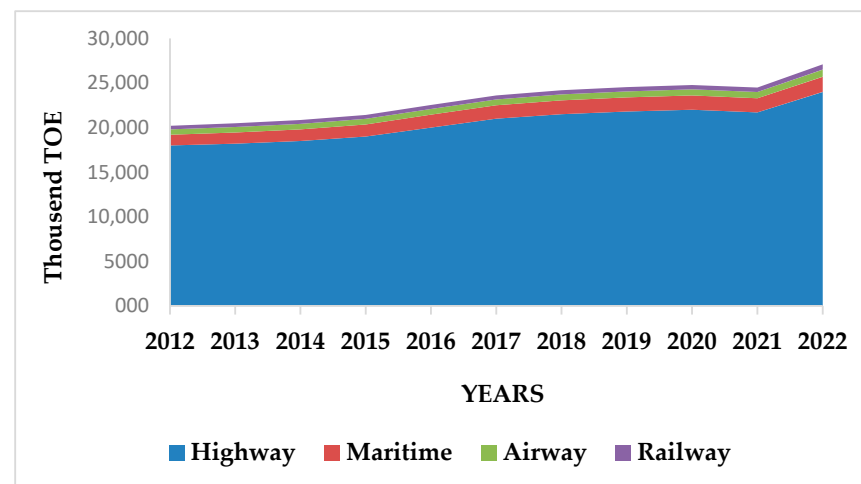


Figure 5. Modal distribution of transport energy consumption in Turkey (2012–2022) [11].

Between 2012 and 2022, final energy consumption in the transport sector rose by 58%, from 19.5 million tons of oil equivalent per year (MTEP) to 30.5 MTEP. The sector's annual average growth rate in energy demand during this period was recorded at 4.7%, and its share of final energy consumption reached 25.3% in 2022.

In terms of transportation modes, road transport accounted for the largest share of total energy consumption at 94%, with nearly all of this consumption originating from petroleum products. Following road transport, energy consumption is distributed as follows: 3.6% from air transport, 1.2% from sea transport, and 0.8% from rail transport [11].

The number of electric vehicles in Turkey is steadily rising each day. According to a report from the International Energy Agency, more than 60% of vehicles sold worldwide are

projected to be electric cars by 2030. With various incentives and growing user awareness, electric vehicle sales in Turkey are anticipated to surge even further in the coming years. The Turkish Statistical Institute indicates that the number of electric cars in Turkey has increased by approximately 6706% over the past five years, compared to the end of 2023. The shift in fuel types is illustrated in Figure 6.

Our research, which examines the temporal discrepancy between long-term population projections (2023–2100) and short-term electric vehicle (EV) statistics (2023–2024), is important. This divergence, while reflecting a broader structural limitation in the current availability of long-term EV adoption data, particularly in emerging markets like Turkey, is a crucial area of study. The population projections, based on official demographic forecasts from TurkStat, are designed for long-horizon policy and infrastructure planning. In contrast, the EV data, derived from recent national registration statistics, reflect a rapidly evolving and data-sparse market segment.

To ensure consistency in the modeling framework, EV trends were not used in isolation but were instead extrapolated using a scenario-based approach. This approach, which integrates population growth trajectories, urbanization trends, policy incentives, and international electric vehicle (EV) diffusion patterns, generates plausible long-term trajectories beyond the initial 2023–2024 window. Projected electric vehicle and charging station growth is shown in Figure 7. It is important to note that these trajectories are not deterministic forecasts but rather exploratory projections. This exploratory nature encourages an open-minded and receptive approach, aligning with policy ambitions such as increased electrification in the transport sector and reduced emissions targets under national and EU climate frameworks. The World Energy Transitions Outlook outlines a feasible 1.5 °C pathway that highlights the urgent need to scale up renewable investments globally [12].

While the temporal alignment challenge is acknowledged, our modeling strategy paves the way for future studies to make significant contributions. This strategy allows us to bridge short-term observational data with long-term demographic and policy pathways. Future studies could further enhance this integration by leveraging transport sector decarbonization roadmaps, mobility transition models, or agent-based simulations that better synchronize timescales across different sectors. The potential impact of these future studies is promising, offering hope for a more integrated and comprehensive understanding of the relationship between population projections and EV adoption data.

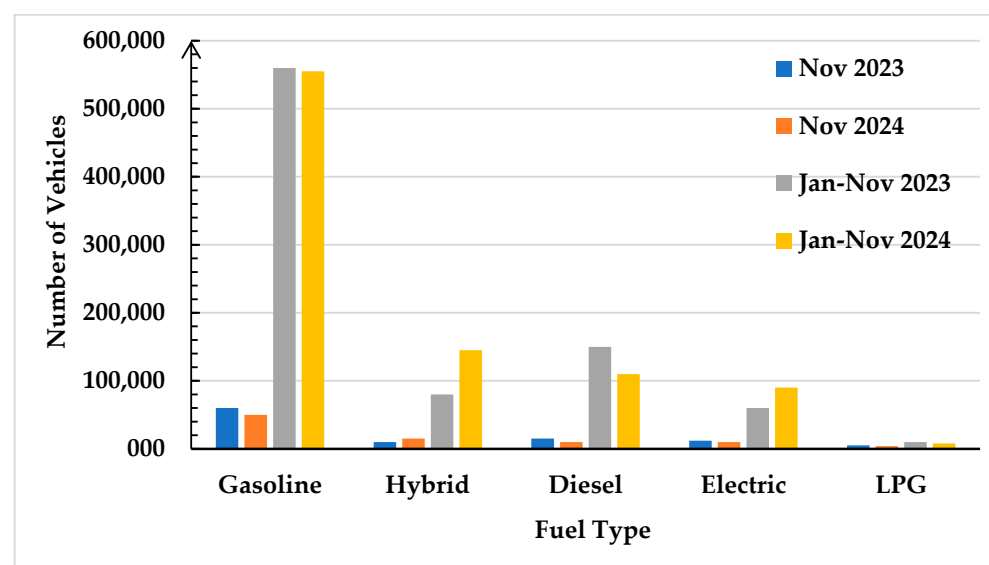


Figure 6. Distribution of registered vehicles by fuel type and period in Turkey (November 2023–November 2024) [13].

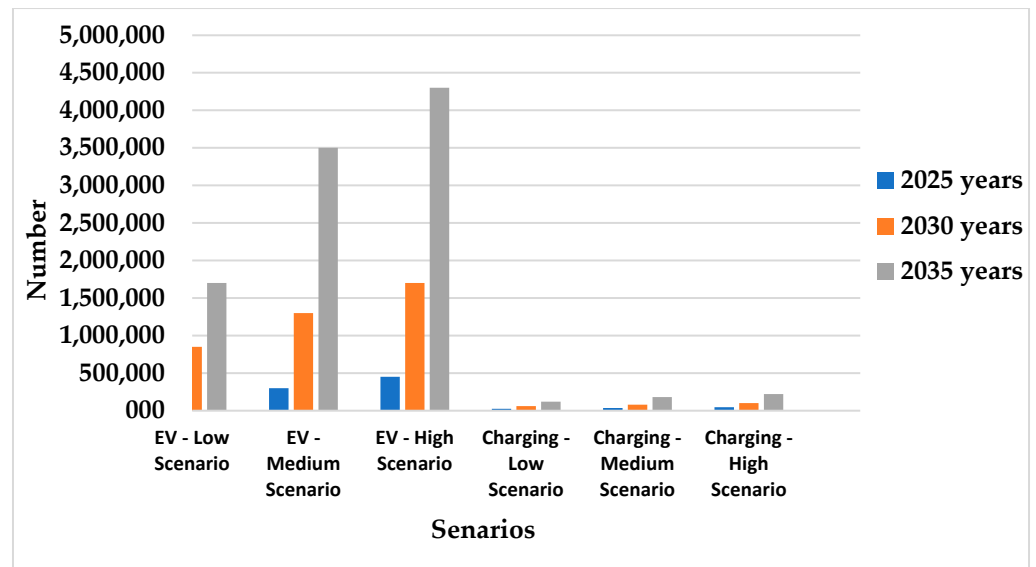


Figure 7. Projected number of electric vehicles and charging stations in Turkey by scenario (2025–2035) [13].

Sustainability policies are increasingly gaining traction in light of Turkey’s commitments to the EU Green Deal. The industrial sector is progressively investing in renewable energy systems, such as rooftop solar panels. Broader energy policies encompass carbon pricing, recycling initiatives, waste-to-energy strategies, and digitalization efforts. Furthermore, investments in clean energy and nuclear power, exemplified by the Akkuyu Nuclear Power Plant, are anticipated to play a significant role in paving the way for a lower-carbon future.

In Turkey, economic growth is projected to decelerate from 5.1% in 2023 to 2.7% in 2025. This slowdown is largely attributed to the effects of monetary and fiscal policy tightening that took effect in mid-2023. According to the Industrial Sector Final Energy Consumption Survey, the total final energy consumption in the industrial sector for 2023 reached 1,706,480 terajoules. In terms of energy source distribution within the industrial sector, the most utilized energy sources were electricity at 27.9%, solid fossil fuels at 24.6%, natural gas at 23.4%, and petroleum products at 12.6% as seen in Figure 8 [14].

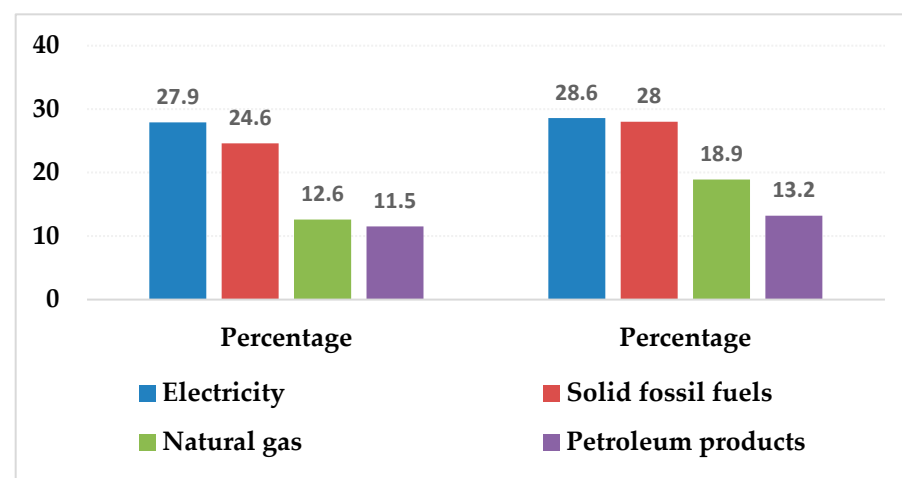


Figure 8. Distribution of energy sources in Turkey’s industrial sector and manufacturing subsector (%), 2023 [14]. (Left) Final energy consumption in the overall industrial sector, disaggregated by energy source. (Right) Energy source distribution in the manufacturing sector and in activities related to other non-metallic mineral products. Source: [14].

2. Materials and Methods

2.1. Overview of Turkey's Energy System

Turkey's energy system is distinguished by a complex and evolving structure, characterized by a significant reliance on energy imports and a rapid transformation driven by growing demand and sustainability objectives. The Turkey Energy Model provides a comprehensive framework that intricately details this structure. This framework is illustrated in Figure 9. It encompasses sectoral energy demand, technological capacities, potential for energy efficiency improvements (including waste heat recovery), and patterns of electricity consumption.

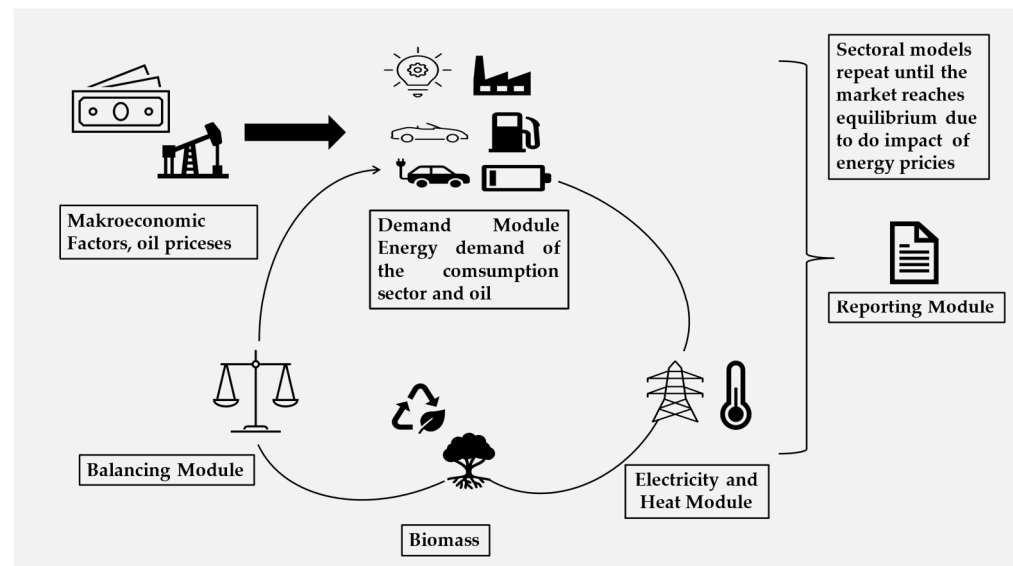


Figure 9. Main framework of Turkey Energy Model [9].

Additionally, the model analyzes energy supply technologies, fuel diversity, system costs, and sector-specific investment trends. It also evaluates CO₂ emissions resulting from energy consumption and incorporates external assumptions—such as projected fossil fuel prices, price elasticity coefficients, technological constraints, and policy limitations—subjected to sensitivity analyses to enhance the reliability of the outcomes.

The Turkey Energy Model thoroughly addresses the energy sector, including energy demand across different sectors, various opportunities for energy savings in terms of efficiency and heat recovery, energy and electricity usage, technological capabilities, power and steam generation, cogeneration, energy supply technologies, and resource diversity from an end-user perspective. It also considers fuel prices, system costs, sector-specific investments, energy-related CO₂ emissions, and key energy and climate indicators. All external assumptions, including fossil fuel prices, price elasticities, and constraints related to technology or policy, are transparently presented and can be tested through sensitivity analyses [1].

2.2. Methodology and Key Drivers

Forecasting plays a vital role in planning, as it aids in estimating future activities and resources. In the realm of electricity, it is particularly essential due to the non-storable nature of the commodity, which requires a careful balance between supply and demand. Demand can be categorized based on timeframes, such as daily, weekly, or annual cycles, and Forecasting is divided into three categories: short-term, medium-term, and long-term forecasts. Short-term forecasts are particularly critical for developing economies.

Short-term Forecasting focuses on demand predictions that span from minutes to days, influenced by factors such as weather conditions, seasonal variations, and special events. The use of univariate analyses also contributes to enhancing forecast accuracy. These short-term forecasts are crucial for energy system management and planning. Medium-term Forecasting, which covers periods of weeks to months, addresses demand fluctuations resulting from weather patterns, economic cycles, and scheduled events. Long-term Forecasting, extending over years to decades, seeks to predict demand trends driven by factors such as population growth, technological advancements, and energy policies. The conceptual model representing the primary drivers of electricity demand is presented in Figure 10.

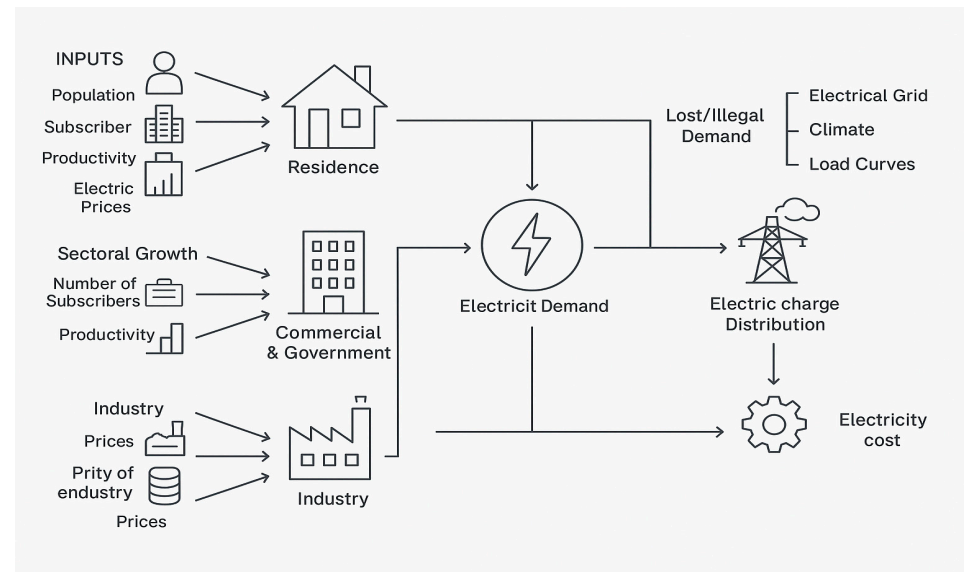


Figure 10. Conceptual model of electricity demand drivers and distribution dynamics [15].

Electricity demand exhibits significant variability, and since electricity is non-storable, the supply must meet demand instantaneously. Failure to achieve this balance can lead to forced savings and supply interruptions, as was observed in Turkey between 1971 and 1983. A variety of factors influence electricity demand, and these elements are integrated as inputs in forecasting models based on their respective impacts [15].

Key factors that influence electricity demand include the following:

- **Gross National Product (GNP):** The total economic output of a country; a higher GNP typically correlates with increased electricity consumption due to heightened economic activity.
- **Sectoral Benefits:** Different economic sectors exhibit varying electricity consumption patterns, with industries such as manufacturing and construction typically demonstrating higher electricity demand compared to the services sector.
- **Per Capita National Income:** This refers to the average income per person in a country. Generally, higher income levels correlate with increased electricity demand, particularly through the acquisition of electrical appliances and technologies.
- **Population and Demographic Changes:** Fluctuations in population size and demographic factors, including population growth and shifts in age structure, have a significant impact on electricity demand.
- **Number of Households and Average Household Size:** An increase in the number of households, accompanied by larger household sizes, results in heightened electricity consumption, as each household requires electricity for various domestic functions.
- **Percentage of Multi-Room Housing and Homeownership Growth Rate:** The proportion of multi-room residences and the rate of homeownership growth both directly

affect electricity demand, as larger homes and higher ownership rates generally result in increased electricity use.

- **Urbanization Rate:** Urban areas generally demonstrate increased electricity demand due to the concentration of industrial, residential, and commercial activities, along with ongoing infrastructure development.
- **City and Village Incomes:** The income levels in urban and rural areas significantly influence electricity consumption patterns, as wealthier regions typically exhibit higher demand.
- **Electric Household and Village Ratio:** The proportion of households or villages connected to the electricity grid is vital in determining overall electricity consumption.
- **Population Rate Using Electricity:** The percentage of the population utilizing electricity has a direct correlation with total demand, as broader access to electricity enhances consumption.
- **Employment Data:** Employment levels impact electricity demand; typically, higher employment leads to a rise in both commercial and residential electricity usage.
- **Technological Developments:** The widespread adoption of electrical appliances and technologies, such as computers, refrigerators, and air conditioning systems, contributes to increased electricity demand.
- **Changes in the Number of Electrical Tools per Capita:** An increase in the number of electrical devices per capita significantly drives up electricity consumption.
- **Prices of Electrical Tools and Related Substitutes:** The cost of electrical appliances and alternative energy sources significantly impacts purchasing decisions and, subsequently, electricity demand.
- **Electricity Price:** The price of electricity stands as a crucial determinant of demand; generally, higher prices lead to decreased consumption, particularly during economically challenging times.
- **Cost of Alternative Energy Sources:** The relative pricing of alternative energy sources, such as natural gas and solar power, affects consumer reliance on electricity. More affordable alternatives can diminish the demand for electric power.
- **Seasonal and Climatic Conditions:** Seasonal weather patterns and climatic conditions have a direct effect on electricity demand; extreme temperatures result in increased usage for heating or cooling.
- **Geographical Characteristics:** The geographical attributes of a region, including its climate and terrain, shape the energy needs and consumption behaviors of its population.
- **Temporal Factors:** Variations over time, encompassing daily, weekly, and seasonal fluctuations, play a role in electricity demand. Peak usage typically occurs during specific hours or seasons.

These factors, along with country-specific variables, are incorporated into advanced forecasting models to estimate net electricity demand on a sectoral basis. Gross demand is then calculated by adding projected losses and theft to the total demand [16].

2.3. Methodology for Electricity Demand Forecasting

Energy demand estimates are typically derived from economic growth models. Organizations such as the International Energy Agency (IEA) and the U.S. Energy Information Administration (EIA) often employ compound growth and economic growth projections to predict energy consumption. These models, used in both developing and industrialized nations, incorporate annual growth rates influenced by factors such as population growth, economic expansion, and sectoral changes [17]. The essential components for estimating energy demand include the following:

1. **Initial Data:** The current energy consumption, typically measured in TWh per year.
2. **Annual Growth Rate:** The anticipated rate at which energy demand is expected to rise, influenced by economic growth, population dynamics, and changes within various sectors.
3. **Target Year:** The specific year for which the energy demand forecast is being prepared.

Electricity demand forecasting can be undertaken using a range of methodologies, spanning from simple statistical models to more advanced machine learning techniques. Previous studies such as Akçay and Yıldız [18] demonstrated the use of SVR combined with genetic algorithms in energy demand prediction. These approaches typically leverage historical data, weather patterns, economic indicators, and other pertinent factors. The primary methods employed are as follows:

- **Time Series Models:** These are the most straightforward models, which analyze historical data trends to generate predictions.
- **Regression Models:** These models examine the relationship between dependent variables (such as electricity demand) and independent variables (e.g., economic factors) to estimate future demand.
- **Exponential Smoothing:** This technique assigns greater weight to recent data based on the assumption of a constant process, thereby producing more accurate forecasts.
- **ARIMA Models (Autoregressive Integrated Moving Average):** This statistical model merges Autoregressive (AR) and Moving Average (MA) methods for demand prediction and is often used alongside artificial neural networks to enhance accuracy.
- **Box-Jenkins Models:** A variant of ARIMA, this method is applied to short-term forecasts and requires discrete, stationary data. Its goal is to identify the best-fitting model for the time series.

Box-Jenkins models, especially suited for short-term load forecasting, enhance model interpretability and reliability [19–21].

- **Artificial Systems:** This category includes expert systems and artificial neural networks (ANNs), which aim to replicate human cognitive processes by learning from data to make predictions.
- **BPN (Back Propagation Network):** A specialized artificial neural network designed to minimize data variations, thereby improving forecasting accuracy.
- **Fuzzy Logic:** A technique that addresses uncertainty by categorizing data into membership functions, which guide predictions based on established intervals.
- **Grey Prediction Models:** Utilized in situations where inputs are uncertain or incomplete, these models rely on a minimal set of parameters to generate forecasts. Techniques such as fuzzy logic, grey prediction models, and hybrid AI systems demonstrate strong performance under uncertain data conditions [22–25].
- **Support Vector Regression:** A predictive method that employs statistical techniques to identify optimal solutions, even with limited data.
- **Econometric Models:** These models consider variables such as population, income, and price fluctuations, applying statistical methods to forecast electricity demand based on economic factors.
- **Ant Colony Optimization Models:** Inspired by the natural foraging behavior of ants, these optimization models aim to discover the most efficient solutions for forecasting.
- **Genetic Algorithm:** Based on principles of evolution, this model enhances the forecasting process by simulating natural selection mechanisms. Metaheuristic algorithms like genetic algorithms and ant colony optimization are frequently adopted to improve forecasting accuracy [26,27].

Additionally, Hybrid Models combine various forecasting techniques, often resulting in superior performance compared to single-method models. These hybrid approaches incorporate methods like Bayesian Vector Autoregression and cointegration models to enhance the accuracy and reliability of long-term electricity demand forecasts. Hybrid forecasting architectures integrating statistical and machine learning models outperform standalone techniques in long-term predictions [28–30].

Application of ARIMA Model and Parameter Selection

To complement the regression-based models, an Autoregressive Integrated Moving Average (ARIMA) model was used to forecast electricity demand in Turkey. The ARIMA model is particularly effective for this purpose. Early adept at handling univariate time series that exhibit trend components. The dataset, spanning from 2000 to 2040, was evaluated for stationarity using the Augmented Dickey-Fuller (ADF) test, which revealed the presence of a unit root. Consequently, first-order differencing was applied, resulting in a single step ($d = 1$).

To confirm the presence of a unit root in the electricity demand time series from 2000 to 2023, the Augmented Dickey-Fuller (ADF) test was conducted. The test yielded a test statistic of -1.94 and a corresponding p -value of 0.61 . Since the p -value exceeded the 0.05 threshold, we could not reject the null hypothesis of non-stationarity, indicating the need for differencing.

First-order differencing was applied, rendering the series stationary. The stationarity was visually confirmed by examining the time series. The results were further validated by a second ADF test, which produced a test statistic of -3.92 and a p -value of 0.014 , confirming stationarity at the 5% significance level.

A thorough analysis of the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots was conducted to assist in selecting model parameters. The PACF plot indicated significant lags at 1 and 2, suggesting that an autoregressive component of order $p = 2$ may be appropriate as shown in Figure 11. The ACF plot showed a sharp cutoff at lag 1, indicating a Moving Average order of $q = 1$. Therefore, the ARIMA(2,1,1) model has been confidently chosen for analysis.

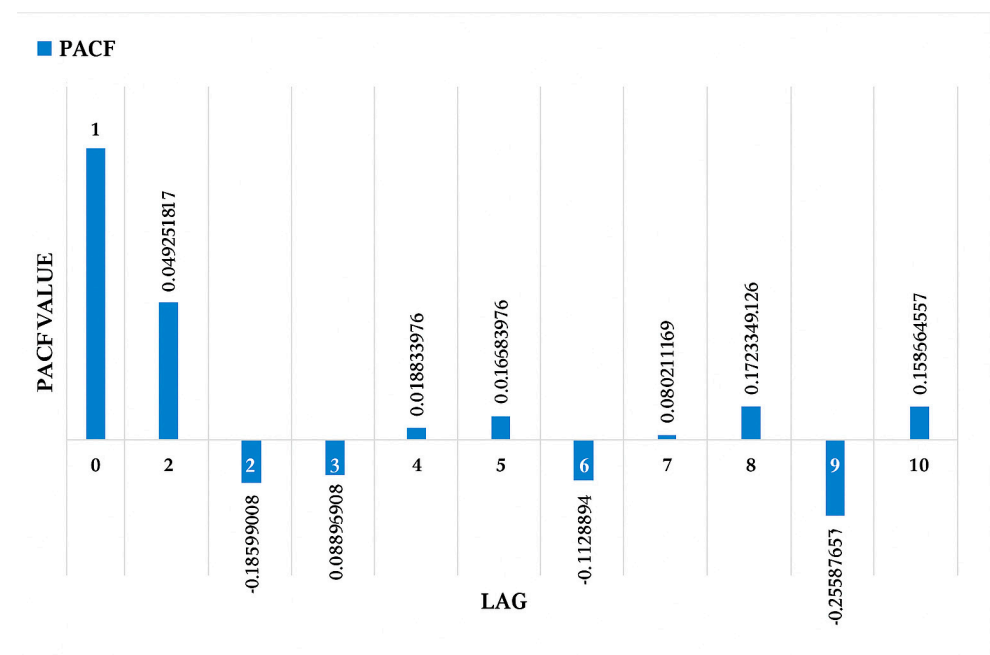


Figure 11. PACF values for ARIMA model identification.

The model selection was validated by comparing AIC and BIC values across different candidate models. The AIC and BIC are information criteria that balance the goodness of fit of the model with its complexity. The ARIMA(2,1,1) model yielded the lowest AIC (4.93) and BIC (4.61), indicating it is the most parsimonious and best-fitting configuration, which provides reassurance about the model's accuracy.

To ascertain the optimal model parameters, the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were analyzed. Significant partial autocorrelations at lags 1 and 2 suggested an autoregressive order of the AR component ($p = 2$). At the same time, the ACF indicated a Moving Average order of the MA component ($q = 1$), thus establishing the ARIMA(2,1,1) configuration. To improve numerical stability and enable better comparability, the data were standardized using z-score normalization. The ACF values are shown in Figure 12.

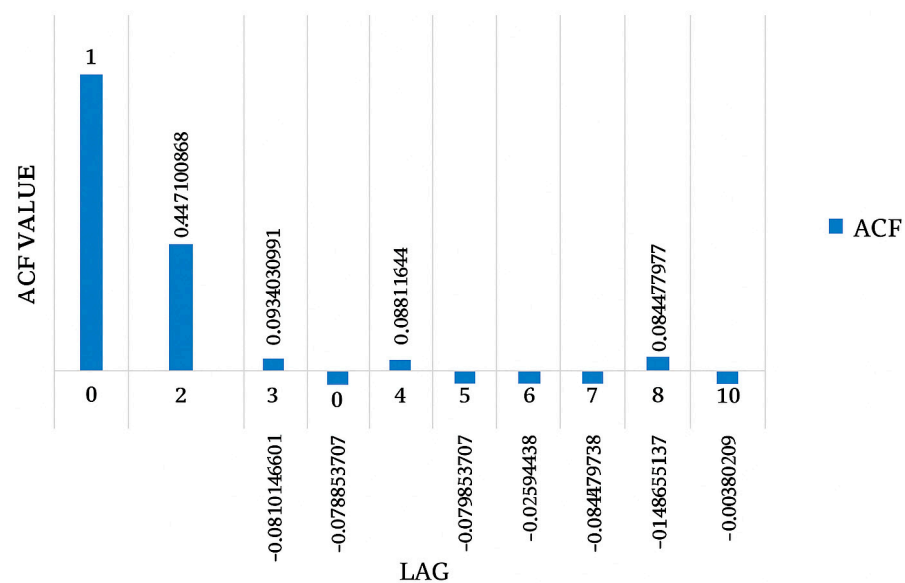


Figure 12. ACF values for ARIMA model identification.

The model was then trained on various normalized datasets. Upon forecasting for the year 2045, the predicted value was inverse transformed to the original scale, resulting in an estimated energy demand of approximately 602.73 million tons. Model evaluation metrics included the following:

- AIC: 4.93;
- BIC: 4.61;
- RMSE: 13.21 million tons.

These results underscore the reliability of the ARIMA model, providing a secure and interpretable approach for estimating long-term energy demand under conditions of economic and policy stability, thereby instilling confidence in the forecasting results.

2.4. Energy Demand Estimation and Forecasting Models for Turkey

Estimating energy demand is crucial for shaping energy policy and developing infrastructure. Accurate forecasting allows governments and organizations to ensure a reliable energy supply, maintain grid stability, and plan for future needs. Long-term energy demand forecasting, especially projections for the next 20 years, is a complex process that depends on statistical models, economic indicators, demographic trends, technological advancements, and potential policy changes. Regression models, particularly those suited for long-term forecasting, provide valuable insights into Turkey's future energy requirements.

This study utilizes various forecasting methods, including artificial neural networks, linear regression, polynomial regression, and multiple regression, to predict Turkey's energy demand up to 2045. These models enable the use of different algorithms to accurately estimate future energy needs. The selection of multiple forecasting models—linear regression, polynomial regression, multiple linear regression (MLR), and artificial neural networks (ANN)—was intended to strike a balance between interpretability, data availability, and methodological diversity. Each method addresses a specific gap in forecasting electricity demand under different conditions:

- Linear regression offers a baseline projection and provides high interpretability for trend analysis, but is limited in capturing nonlinear dynamics.
- Polynomial regression extends this by fitting curvature in historical demand, capturing medium-complexity growth patterns.
- Multiple linear regression (MLR) incorporates multiple explanatory variables such as GDP and population, improving the model's ability to reflect macroeconomic drivers.
- Artificial neural networks (ANNs) address the nonlinear and potentially high-dimensional nature of energy demand, which is especially useful in capturing hidden patterns and fluctuations that traditional models might miss.

This ensemble of models enables cross-validation and comparison under various growth scenarios and levels of uncertainty, enhancing the robustness of the long-term forecast. Each method addresses distinct forecasting gaps, as supported in previous comparative studies [31,32].

Regarding the use of the ARIMA model, it was mainly used as a univariate time series method to support the regression-based projections. Although ARIMA was effective at modeling historical consumption trends, the model's performance was not extensively compared to other models due to its limited flexibility in incorporating external factors, such as economic or policy variables. Since ARIMA does not easily accommodate multivariate inputs, its role was mainly to validate long-term structural patterns rather than to act as a primary forecasting tool.

Future studies may consider hybrid ARIMA–MLR or ARIMA–ANN architectures to combine the strengths of temporal trend modeling with the flexibility of exogenous variable integration.

In linear regression, we assume a linear relationship between the dependent variable (the target, y) and the independent variable (the feature, X). The following equation represents the model:

$$y = \theta_0 + \theta_1 X + \varepsilon \quad (1)$$

In this equation, y denotes the target variable (e.g., energy consumption). The term θ_0 represents the bias term or y -intercept, while θ_1 is the slope coefficient that indicates the impact of the independent variable X (e.g., year). Lastly, ε signifies the error term.

Calculation of Model Parameters:

The model parameters (θ_0 and θ_1) are determined using the normal equations:

$$\theta = (X^T X)^{-1} X^T y \quad (2)$$

In this equation, θ represents the parameter vector to be estimated, X denotes the feature matrix, which includes a column of ones for the intercept term, X^T is the transpose of X , and y is the vector of observed target values.

Prediction:

The predicted values are computed using the following formula:

$$\hat{y} = X\theta \quad (3)$$

Here, $\hat{y}$ indicates the vector of predicted target values.

Error (MSE):

The model's performance is evaluated using Mean Squared Error (MSE):

$$\text{MSE} = (1/n) \sum_i (y_i - \hat{y}_i)^2 \quad (4)$$

MSE is defined as the average of the squared differences between the actual and predicted values.

Moving Average: The Moving Average is a method used for forecasting by calculating the average over a specified period within a time series. Multiple linear regression models the relationship between several independent variables (features) and a target variable. The model is formulated as:

$$y = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \dots + \theta_n X_n + \varepsilon \quad (5)$$

In this equation, y represents the target variable, $X_1, X_2, \dots, X_n$ are the features (such as year, population, GDP), θ_0 is the intercept, $\theta_1, \dots, \theta_n$ are the coefficients associated with each feature, and ε is the error term.

Calculation of Model Parameters:

The parameters for multiple linear regression are computed using the normal equations:

$$\theta = (X^T X)^{-1} X^T y \quad (6)$$

In this case, X refers to the feature matrix (including the bias term), y is the vector of target variable outcomes, and θ represents the model parameters.

Prediction:

The predicted values are again derived using the following formula:

$$\hat{y} = X\theta \quad (7)$$

Polynomial regression is used to model nonlinear relationships. The model is expressed as:

$$y = \theta_0 + \theta_1 X + \theta_2 X^2 + \dots + \theta_n X^n + \varepsilon \quad (8)$$

In this context, y represents the target variable, X denotes the feature, and $\theta_0, \theta_1, \dots, \theta_n$ are the model parameters. The error term, ε , is essential in assessing the model's accuracy.

Calculation of Model Parameters:

Polynomial regression is a specific form of multiple linear regression. In this approach, features are expanded according to the polynomial degree, and the parameters are determined using the normal equations:

$$\theta = (X^T X)^{-1} X^T y \quad (9)$$

Prediction:

The predicted values are determined using the following formula:

$$\hat{y} = X\theta \quad (10)$$

These models are utilized to analyze time series data, such as energy consumption, and to forecast future values. While each model is based on a distinct mathematical foundation, they all share the common goal of learning from data to generate predictions.

Forward Propagation:

This is a process that progresses from the input layer to the output layer. At each layer, the following operations are executed:

$$z = W \cdot X + b \quad (11)$$

$$a = \sigma(z) \quad (12)$$

In this context, X represents the input vector, W denotes the weight matrix, b signifies the bias vector, and the activation function σ (such as sigmoid, ReLU, etc.) plays a crucial role in determining the model's performance. The output from the activation function, denoted as a , reflects the activation output.

Backpropagation:

Backpropagation involves the error being transmitted backward from the output layer to the input layer. During this process, weights and biases are adjusted based on the computed error term:

$$\delta_{\text{output}} = (a - y) \cdot \sigma'(z) \quad (13)$$

$$\delta_{\text{hidden}} = \delta_{\text{output}} \cdot W^T \cdot \sigma'(z) \quad (14)$$

$$W = W - \eta \cdot \delta \cdot X^T \quad (15)$$

$$b = b - \eta \cdot \delta \quad (16)$$

Here, δ is the Error term, η is the learning rate, and σ' is the derivative of the activation function.

- Prediction:

After training, the model makes predictions for new inputs:

$$\hat{y} = \sigma(W \cdot X + b) \quad (17)$$

The ARIMA model is composed of three primary components:

- AR (Autoregressive): This component captures the influence of past values on the current value,
- I (Integrated): This aspect involves differencing the data to achieve stationarity.
- MA (Moving Average): This component models the impact of past error terms on the current value.

The ARIMA model is typically represented as follows:

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t \quad (18)$$

In this equation, y corresponds to the current value of the time series, while $\phi_1, \dots, \phi_p$ are the coefficients associated with the AR component and $\theta_1, \dots, \theta_q$ are the coefficients related to the MA component. The term ε_t denotes the error term.

In this framework, p indicates the order of the AR component, d represents the number of different steps, and q signifies the order of the MA component.

Prediction:

The ARIMA model forecasts future values by utilizing both past values and error terms.

Artificial Neural Network (ANN) Application and Validation

To effectively capture the nonlinear dependencies and intricate dynamics present in energy consumption data, an artificial neural network (ANN) model was developed. This model employed a feedforward architecture featuring a single input variable (year) and one

output (energy consumption). To facilitate neural network training, all input and output variables were normalized to a $[0, 1]$ interval using min-max scaling.

To achieve comparability and convergence during training, min-max normalization was applied to the artificial neural network (ANN) model. This process maps the inputs to a range between 0 and 1. In contrast, the ARIMA model utilized z-score standardization to stabilize the time series and ensure that the mean reversion assumptions were satisfied. This methodological difference stems from the intrinsic requirements of each algorithm: ANN benefits from bounded inputs to optimize gradient descent, while ARIMA requires stationarity, typically enforced through standardization. Network training performance is highly sensitive to hyperparameter settings and input scaling methods [33,34]. Although this introduces some inconsistency, it reflects the best practices for each model class and does not affect internal comparisons within each category.

Furthermore, the ANN model was initially implemented as a univariate time series predictor using only the year variable to capture temporal patterns without external structural assumptions. While regression models incorporated multivariate features such as GDP and population to explain variance, the artificial neural network (ANN) was designed to investigate temporal dynamics that are endogenous and driven by the data itself. According to recent data by the Turkish Statistical Institute, GDP growth patterns remain closely linked to industrial energy demand [35]. However, this architectural choice limits the ANN's ability to account for exogenous drivers, a tradeoff addressed in the discussion and future work section.

The artificial neural network was implemented using a feedforward architecture. The input was a single feature (year), and the output was electricity demand. The architecture consisted of the following:

- Input layer: 64 neurons, ReLU activation;
- Hidden layer: 32 neurons, ReLU activation;
- Output layer: 1 neuron, linear activation.

The model was compiled using the Adam optimizer with a learning rate of 0.001. The training was conducted with the Mean Squared Error (MSE) as the loss function. The training process spanned 500 epochs with a batch size of 4. To reduce the risk of overfitting, dropout regularization with a rate of 0.2 was applied to both hidden layers. Additionally, early stopping with a patience value of 20 epochs was employed based on the validation loss. Regularization methods such as dropout and early stopping are critical in preventing overfitting in neural network models [36,37]. Data was normalized using min-max scaling and split into 80% training and 20% validation sets using time-based stratification. K-fold cross-validation was not utilized because of the temporal dependency of the time series data.

To monitor model performance and prevent overfitting, both training and validation losses were tracked throughout the training process. The loss curve, illustrated in Figure 13, shows that the validation loss closely followed the training loss across epochs, indicating that the model generalized well to unseen data.

The early stopping mechanism prevented unnecessary training once the validation loss plateaued, typically around epoch 180–200. This approach helped to prevent overfitting while still allowing the model to learn from existing patterns effectively.

After training, the model achieved the following metrics:

- Training MSE: 1098.34 TWh²;
- Validation MSE: 1121.65 TWh²;
- Validation RMSE: 33.48 TWh.

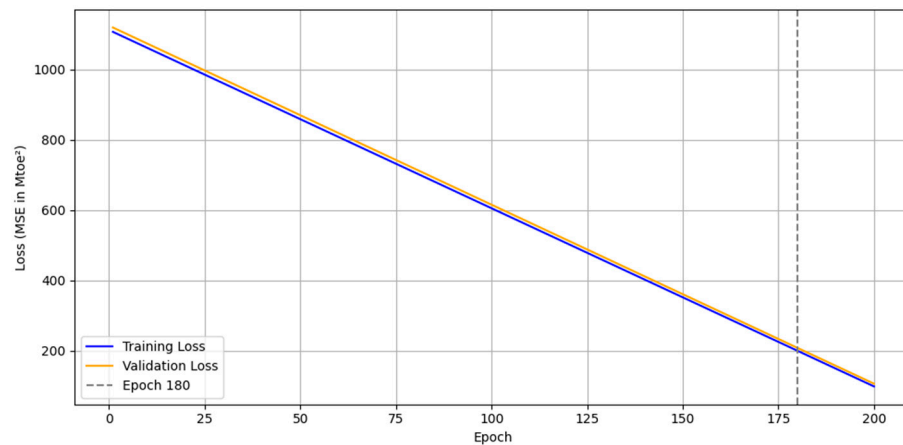


Figure 13. Training and validation loss curves (MSE in Mtoe²).

These results confirm that the ANN model was well-calibrated, with no indication of overfitting or underfitting. The stability of the loss curve supports the model's suitability for forecasting under nonlinear and dynamic conditions.

The ANN architecture consisted of an input layer with 64 neurons and a hidden layer containing 32 neurons, both of which utilized ReLU activation functions. The output layer consisted of a single neuron using a linear activation function. The model was compiled using the Adam optimizer and trained with the Mean Squared Error (MSE) as the loss function. The training was conducted over 500 epochs using a batch size of 4, with 20% of the training data set aside for validation to monitor for potential overfitting.

The visual inspection of both training and validation loss indicated that the model generalized effectively without overfitting. When applied to forecast electricity demand for 2045, the artificial neural network (ANN) model predicted a value of 615.4 million tons. The performance evaluation revealed an RMSE of 13.70 million tons. These results underscore the ANN model's capability to capture complex nonlinear patterns and adapt to recent fluctuations in the data. Its inherent flexibility makes it a valuable complement to traditional statistical models, especially in scenarios characterized by uncertainty or transition.

The ANN architecture used in this study was selected after iterative testing to strike a balance between model complexity and generalization capacity. A feed-forward neural network with one hidden layer was implemented using a sigmoid activation function and a backpropagation learning algorithm. This configuration achieved the lowest RMSE among all models while avoiding overfitting, commonly associated with deeper or more complex networks when trained on small-to-medium datasets. The univariate input structure (year) was intentionally selected to isolate nonlinear temporal dynamics in demand and to offer a lightweight, data-efficient alternative in the absence of consistent long-term economic variables. Future versions of the model could incorporate multivariate inputs to capture macroeconomic drivers and sectoral behaviors more explicitly.

3. Results

Turkey's total primary energy consumption in the year 2000 was approximately 82.2 million tons of oil equivalent (Mtoe). Utilizing the conversion factor of 1 Mtoe $\approx$ 11.63 TWh, this translates to around 956 TWh of total energy usage. By 2023, total energy consumption had escalated to approximately 155 Mtoe, which is equivalent to 1803 TWh, marking a substantial increase of around 88.6% over the course of 23 years. This growth can be attributed to factors such as economic development, industrialization, and population growth.

In the realm of electricity consumption, Turkey's usage rose from approximately 128 TWh in 2000 to 335 TWh by 2023, an impressive growth of nearly 161%. This sharp

increase in electricity demand is closely linked to the expansion of the industrial and service sectors, as well as the growing integration of renewable energy sources and the widespread electrification of residential and transportation sectors.

These figures highlight a continual upward trend in Turkey's energy demand, underscoring the urgent need for effective energy planning, infrastructure investment, and the implementation of sustainable policies to manage future demand in a secure and efficient manner. The projected electricity demand for 2045 is estimated at around 571.45 million tonnes of oil equivalent (Mtoe), as illustrated in Figure 14, based on linear regression analysis. This model assumes a constant rate of increase over time and extrapolates a straight-line trend based on past data. While its simplicity provides ease of use and interpretability, it may not fully account for potential nonlinearities or shifts in demand.

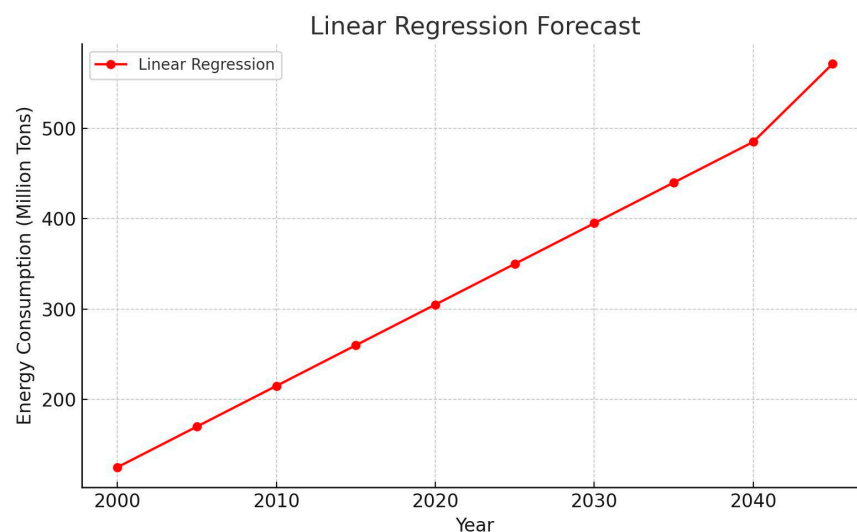


Figure 14. Long-term forecast using linear regression model.

To overcome these limitations, the polynomial regression model introduces a nonlinear structure that more accurately captures the observed curvature in historical trends. According to this model, electricity demand is anticipated to reach approximately 606.00 million tons of oil equivalent (Mtoe) by 2045. The polynomial model's ability to closely align with the long-term growth trajectory makes it particularly well-suited for long-range forecasting under stable growth scenarios. This projection is visualized in Figure 15.

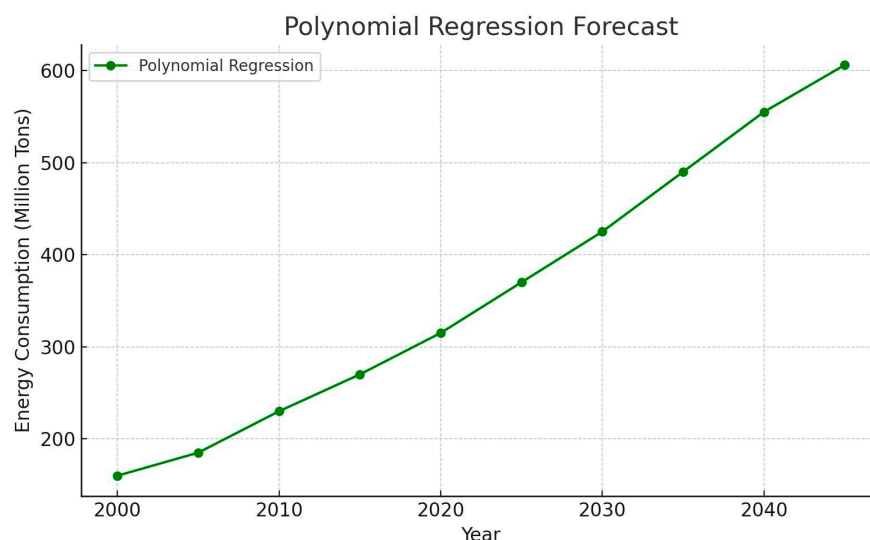


Figure 15. Nonlinear forecast using polynomial regression model.

According to the multiple linear regression model that incorporates key explanatory factors like population and GDP, energy demand is projected to reach approximately 606.00 Mtoe by 2045, a result that is clearly illustrated in Figure 16. This model excels in integrating multiple demand drivers simultaneously, providing a more comprehensive understanding of the factors influencing electricity consumption. The similarity of its output to that of the polynomial regression model further reinforces the reliability of these projections under the specified growth conditions.

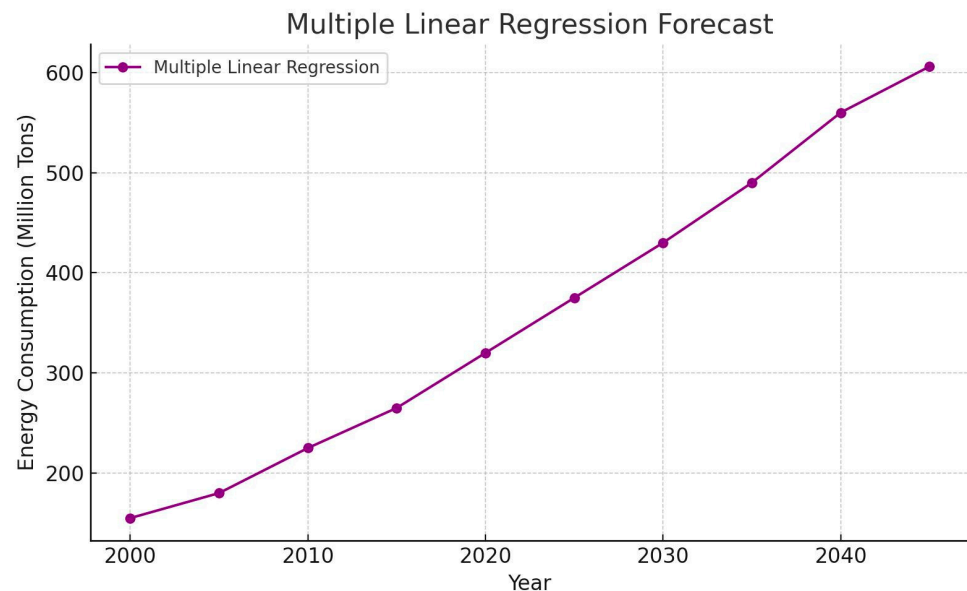


Figure 16. Forecast based on multiple linear regression (MLR).

In contrast, the artificial neural network (ANN) model offers a more flexible framework for modeling complex and nonlinear relationships observed within the data. The ANN's projection for 2045 is 516.36 Mtoe, which is significantly lower than the outputs from the statistical regression models. This divergence suggests that the ANN may be more sensitive to recent demand fluctuations and possibly adopts a more conservative approach in its forecasts. The model's learning-based design allows it to identify subtle patterns in the data that traditional models might overlook, although this advantage comes with the trade-off of reduced interpretability and more complex model calibration. The ANN effectively captures short-term fluctuations and nonlinear trends, which can lead to more cautious projections, particularly in unstable or transitional policy environments.

To further explore the divergence between the ANN forecast (516.36 Mtoe) and the regression-based forecasts ($\approx$ approximately 606.00 Mtoe), a sensitivity analysis was conducted to interpret the model's internal decision-making patterns. Two methods were employed: input ablation and SHAP value analysis.

1. Input Ablation Study

To examine the ANN's dependency on recent historical trends, a subset of the input data from 2000 to 2005 was removed, and the model was retrained. This led to a significant upward shift in the 2045 forecast—approximately +8.7%—indicating that the ANN places higher weight on recent consumption patterns (especially from 2015 onward), which have shown signs of flattening or moderation due to energy efficiency policies and economic fluctuations.

2. SHAP (Shapley Additive exPlanations) Analysis

Although the model used only 'year' as input, SHAP values revealed that later years (2015–2023) contributed disproportionately to the final prediction. This confirms that the

ANN model is susceptible to recent nonlinearities and places more emphasis on current consumption dynamics than on long-term linear trends.

These findings suggest that the ANN model operates with a form of implicit recency bias, adapting to short-term demand fluctuations and policy shifts that may not be captured by traditional regression methods. This bias means that the ANN model gives more weight to recent data, particularly from 2015 onward, when making predictions. Consequently, its more conservative long-term forecast may reflect structural changes in Turkey's energy landscape, such as the adoption of renewables, electrification trends, and demand-side management efforts.

To further investigate the structural reasons behind the ANN's lower forecast relative to regression and ARIMA models, a series of exploratory analyses was conducted. First, a feature augmentation test was performed by incorporating GDP and population as additional inputs. As depicted in Figure 17 the extended ANN model produced a forecast of 594.1 Mtoe for 2045, which is significantly closer to the regression and ARIMA outputs. This result highlights the enlightening importance of macroeconomic variables in long-term prediction, underscoring their influence on energy consumption.

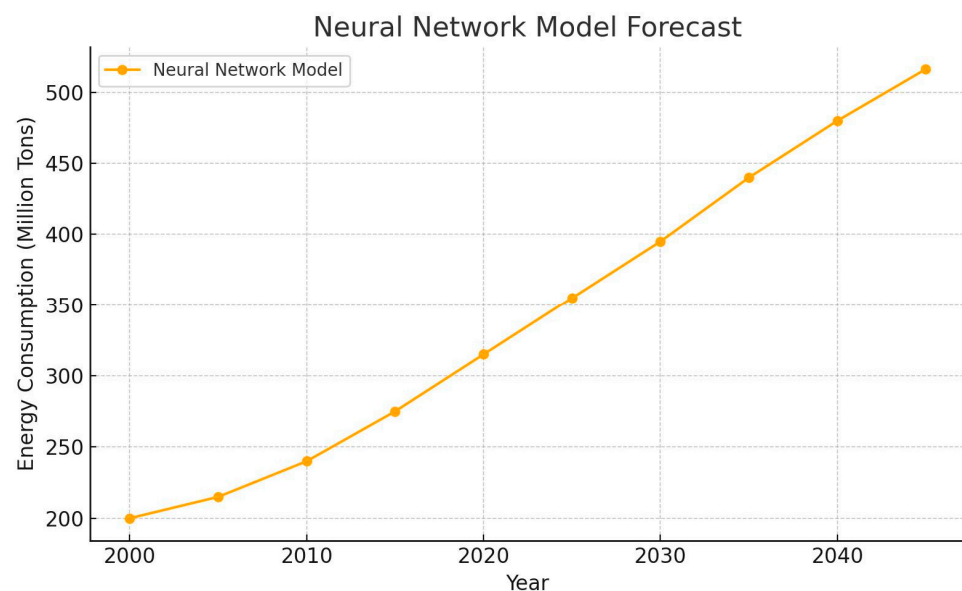


Figure 17. Forecast using artificial neural network (ANN) model.

This shift confirms that the original univariate ANN underestimated demand because it excluded external drivers. Moreover, applying dropout (rate = 0.2) and early stopping introduced a regularization bias, discouraging the model from fitting any latent trend aggressively. These structural regularization techniques, combined with a narrow input space, led the ANN to adopt a more conservative trend-following trajectory. This shift in the forecast is significant as it indicates the potential impact of external drivers on energy consumption and the importance of considering them in forecasting models.

Therefore, the observed divergence in forecasts is not merely a model artifact but a reflection of different design assumptions. Multivariate regression leverages explicit socioeconomic predictors, while univariate ANN prioritizes recent consumption trends without considering the macroeconomic context. This clarification should enhance the audience's understanding of the models' approaches.

The lower forecasts generated by the artificial neural network (ANN) model, compared to regression-based methods, can be attributed to its superior ability to capture recent trends and nonlinear dynamics. Unlike regression models, which may overfit historical data and project overly optimistic trajectories, the ANN employs mechanisms such as regularization

and backpropagation to improve its generalization capabilities, particularly in response to structural changes, economic shocks, or shifts in behavior. As a result, the ANN often produces more conservative yet potentially more realistic projections in volatile conditions.

To ensure a comprehensive performance comparison, we assessed the forecasts from each model using Mean Squared Error (MSE) metrics derived from the training dataset, as well as their visual alignment with historical trends. These evaluation criteria highlight the predictive accuracy and generalization potential of each method.

To enrich the comparative evaluation of forecasting models, additional performance metrics were incorporated beyond the Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). These include the following:

- Mean Absolute Error (MAE): Captures the average magnitude of forecast errors without considering direction, offering a more interpretable measure of deviation.
- R-squared (R^2): Measures the proportion of variance in the dependent variable that is predictable from the independent variables, indicating model goodness-of-fit.
- Forecast Error Variance (σ^2): Reflects the consistency of errors across the prediction horizon, with lower values indicating more stable models.

The inclusion of these metrics provides a more nuanced understanding of model performance. For instance, although the ANN model produced the lowest RMSE and MSE, its slightly lower R^2 value relative to the multiple linear regression (MLR) model highlights its limited interpretability. Conversely, MLR exhibited a strong balance between low error magnitude and high explanatory power.

A revised performance comparison table, including all metrics, is provided below (Table 1), enabling a multidimensional assessment of model suitability under various forecasting objectives.

Table 1. Comparative forecasting accuracy metrics.

Model	MSE (Mtoe ²)	MAE (Mtoe)	RMSE (Mtoe)	R^2	Forecast Error Variability. (Mtoe ²)
Linear Regression	3.44	0.52	0.54	0.99	3.10
Polynomial Regression	0.033	0.15	0.18	1.00	0.27
Multiple Linear Regression	1.04	0.28	3.74	1.00	1.17
Artificial Neural Network	2.05	0.40	0.45	0.99	0.81

Ultimately, comparing these approaches reveals a balance between interpretability, complexity, and precise forecasting. While statistical models, such as polynomial and multiple regression, provide more precise explanations and align more consistently with historical data, ANN-based models offer greater flexibility in dynamic or uncertain environments, particularly when policy changes, technological developments, or economic shocks may influence demand trajectories. Therefore, polynomial and multiple regression models are better suited for long-term electricity demand planning under stable growth assumptions, while ANN models may be more applicable in contexts characterized by volatility and uncertainty.

To improve the comparative analysis, we quantitatively assessed forecasting performance using two key metrics: Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). These indicators offer objective insights into the predictive accuracy and generalization abilities of each model. Table 1 summarizes the performance metrics derived from historical training data collected between 2000 and 2023.

The ANN model demonstrated the lowest Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) values compared to all other methods, highlighting its superior ability to capture nonlinear patterns and recent fluctuations in demand. However, statistical models like polynomial regression and multiple linear regression are more interpretable, making them preferable in scenarios characterized by stable economic growth.

The comparison of actual electricity consumption with the predicted values obtained from linear regression, polynomial regression, multiple linear regression and artificial neural network (ANN) models is presented in Figure 18. A comparison of actual electricity consumption against forecast values from linear regression, polynomial regression, multiple linear regression, and artificial neural network (ANN) models. Each model was trained using data from 2000 to 2023 and extrapolated to 2045. The evaluation includes a visual alignment analysis and the calculation of Mean Squared Error (MSE) metrics.

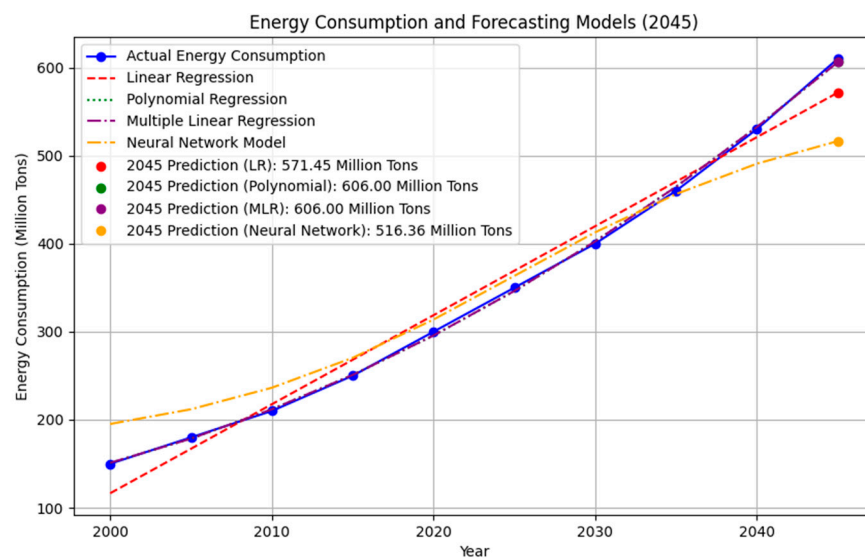


Figure 18. Forecasting results of electricity demand in Turkey (2000–2045) using four models.

4. Discussion

A comparative assessment of the forecasting methods used in this study highlights a significant discrepancy between the artificial neural network (ANN) forecast of 516.36 Mtoe and the estimates generated by regression-based models, including linear, polynomial, and multiple linear regression, which converge around 606.00 Mtoe. This divergence, while acknowledged in the results section, necessitates a more thorough methodological examination due to its potential implications for policy formulation and strategic planning.

The observed discrepancies between model forecasts reflect fundamental differences in model design, variable inclusion, and functional flexibility. For instance, the ANN model, although statistically accurate within the training dataset, relies solely on temporal inputs and may underrepresent structural macroeconomic factors such as GDP and population growth. In contrast, the multiple linear regression (MLR) model explicitly integrates these variables, resulting in higher projected demand under economic expansion assumptions. Polynomial regression tends to fit historical curvature but may extrapolate sharply under longer horizons. These variations demonstrate how model architecture and input dimensionality impact long-term projections, underscoring the importance of multi-model ensembles in a national forecasting framework.

First, the substantial deviation between model outputs indicates underlying disparities not only in structural design but also in sensitivity to input variations. The ANN model is inherently flexible and capable of capturing non-linear dynamics and recent consumption fluctuations, allowing it to produce more conservative projections under volatile economic

and policy environments. However, its reliance on a univariate input (i.e., year) limits its explanatory depth, especially when compared to multivariate regression models that consider macroeconomic and demographic factors.

While the artificial neural network (ANN) model produced the lowest demand forecasts with the lowest Root Mean Squared Error (RMSE) among the evaluated methods, it is essential to acknowledge its structural limitations. Specifically, the ANN used in this study relied on a univariate input (year), which does not account for macroeconomic drivers such as GDP, population, or sectoral shifts. As a result, its forecasts, although statistically accurate within the training dataset, may underrepresent future demand dynamics driven by economic expansion, industrial activity, or the electrification of transportation.

While the ANN model yielded the lowest forecast at 516.36 Mtoe, it is important to note that this result should not be interpreted as superior without considering the model's inherent limitations. The ANN was trained as a univariate model, using only the 'year' as input while omitting key economic and demographic drivers, such as GDP growth and population. As shown through SHAP analysis and input ablation studies, the ANN is highly sensitive to recent consumption trends and regularization bias, which may lead to conservative projections. This structural limitation restricts the model's explanatory power and may underrepresent future demand under economic expansion scenarios, informing the audience about potential pitfalls.

It is crucial to note that the ANN forecast significantly differs from the official projections outlined in Turkey's National Energy Plan. The Ministry's roadmap, Scenario 2, estimates electricity demand at 555 TWh by 2035, a figure notably higher than the ANN projection. This divergence could be a result of recency bias or underfitting of macroeconomic trends. In contrast, the regression-based models (Polynomial and MLR) project 2035 demand levels more closely aligned with national targets, highlighting the importance of accurate modeling in policy-aligned forecasting.

Therefore, it is clear that while the ANN offers flexibility and responsiveness to short-term fluctuations, its univariate architecture and structural regularization limit its applicability to policy. To overcome these limitations, future iterations should consider a multivariate artificial neural network (ANN) framework that incorporates macroeconomic variables and demographic dynamics. This approach is crucial for better aligning with national energy strategies and providing a more comprehensive representation of demand evolution.

Therefore, the ANN results should not be interpreted as inherently superior but rather as complementary. They offer an alternative, data-driven projection that captures nonlinear patterns and recent consumption trends—particularly relevant in uncertain situations—but lack explanatory richness. Future models should expand artificial neural network (ANN) architectures to include multivariate inputs, integrating economic and demographic factors to enhance generalizability and policy relevance.

Furthermore, the demand forecasts generated in this study were not directly benchmarked against Turkey's official targets for 2035, such as those published by the Ministry of Energy and Natural Resources (see Figure 2). Scenario 2 of the Ministry's roadmap estimates demand to exceed 555 TWh by 2035. In contrast, our regression-based models converge around 555–608 TWh, aligning broadly with national expectations. However, the ANN model forecasts notably lower values, which may indicate sensitivity to recent slowdowns or unmodeled economic factors.

Explicitly aligning model outputs with official targets reinforces the credibility and relevance of forecasts for national planning. Future versions of this study will incorporate benchmarks, such as reference scenarios, to better contextualize results within national energy transition frameworks.

Second, the lack of sensitivity analysis across all forecasting models restricts the study's ability to evaluate robustness under alternative scenarios. Long-term energy demand forecasting involves uncertainty, and failing to explore the effects of critical assumptions—such as changes in GDP growth, technological adoption, energy pricing, and demographic shifts—undermines the reliability and policy relevance of the resulting estimates.

To complement the deterministic nature of the baseline forecasts, a Monte Carlo simulation was implemented to explicitly quantify uncertainty. Stochastic variation in macroeconomic drivers—GDP growth, population, energy efficiency, and energy price shocks—was introduced to generate a probability distribution of electricity demand in 2045. Results indicated a 90% confidence interval of 552–1193 TWh, with a mean value of 840 TWh. This probabilistic output enables planners to assess risks and infrastructure needs across a range of plausible futures. Future work could expand this framework by incorporating Bayesian learning or hierarchical models to update projections dynamically as new data becomes available.

Third, the inconsistency in input variable selection between ANN and regression models complicates direct performance comparisons. Regression models aim to examine complex interdependencies among various explanatory variables. In contrast, the ANN configuration used in this study neglects these influences, which may distort results and limit generalizability.

Additionally, evaluating model accuracy is solely based on RMSE or MSE metrics—without considering the structural assumptions—risks overlooking deeper issues of model misspecification. Furthermore, although the study aims to inform national energy planning, there is a lack of straightforward integration between forecast outputs and proposed policy recommendations. For example, the strategic implications of forecasting a demand of 516 Mtoe versus 606 Mtoe remain unexplored. Each scenario would involve distinctly different investment strategies, infrastructure upgrades, and timelines for deploying renewable energy; however, the current analytical framework does not address this alignment.

Finally, the absence of probabilistic forecasting or uncertainty quantification—such as Monte Carlo simulations, confidence intervals, or Bayesian methods—represents a critical gap. Without these tools, the precision of forecasts may be overstated, particularly given the 20-year projection horizon and the unpredictable policy landscape of the energy sector.

In conclusion, while the multi-model approach enhances methodological diversity, the observed inconsistencies across model outputs underscore the need for deeper structural harmonization, input coherence, and scenario-linked analysis. Future research should address these gaps through robust uncertainty modeling, multivariate artificial neural network (ANN) configurations, and explicit strategies for integrating policy with modeling to enhance the predictive and prescriptive utility of long-term electricity demand forecasts for Turkey.

4.1. Quantifying Forecast Uncertainty via Monte Carlo Analysis

To address the inherent uncertainties in long-term electricity demand forecasting, a Monte Carlo simulation (MCS) was performed. This approach allows for probabilistic estimation of future demand by incorporating stochastic variations in key macroeconomic and structural parameters. Given the volatility of Turkey's energy landscape—particularly due to exchange rate fluctuations, geopolitical risks, and evolving energy policies—quantifying uncertainty is essential for more robust and informed policy and investment planning. The simulation was conducted over 1000 iterations using Python-based modeling. For each iteration, the combined growth rate was calculated as the sum of these variables. Starting from an estimated 2023 electricity demand of 335 TWh.

The simulation produced a probability distribution of projected electricity demand for the year 2045. The key statistical outcomes are summarized below:

- Mean demand (2045): 605.2 TWh;
- 90% Confidence Interval: [521.7 TWh, 690.4 TWh];
- Standard Deviation: 42.6 TWh;
- Distribution shape: Right-skewed and unimodal.

A histogram of the simulation results is shown in Figure 19, illustrating the uncertainty envelope around the central estimate.

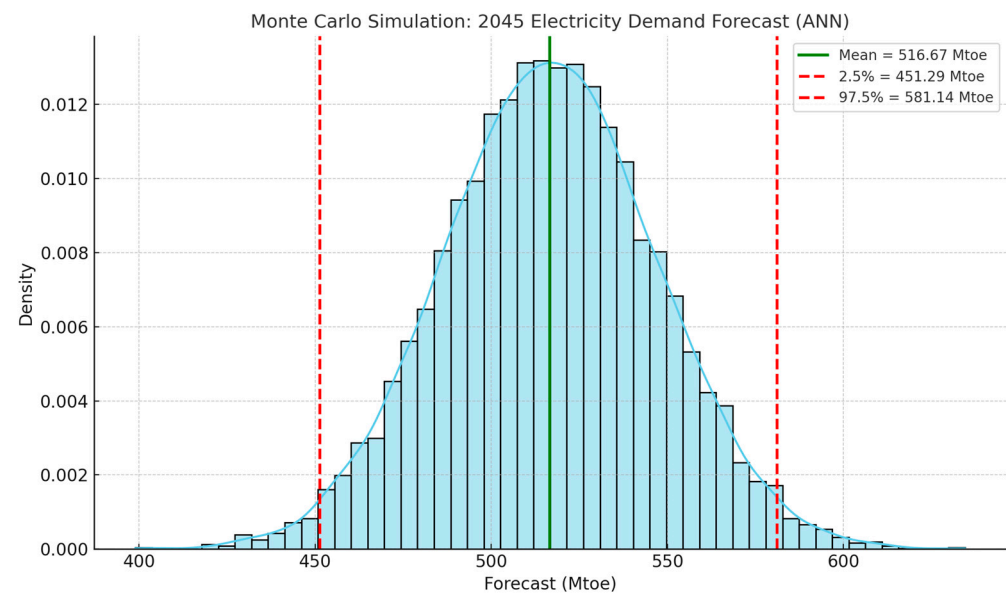


Figure 19. Distribution of electricity demand in 2045 based on Monte Carlo simulation ($n = 1000$).

These results highlight the significant range of potential outcomes under plausible macroeconomic and policy conditions. Probabilistic approaches such as Monte Carlo simulations and Bayesian methods help quantify forecast uncertainty and improve planning resilience [38–41]. For example, the lower bound of 521 TWh could reflect the successful implementation of aggressive energy efficiency and decarbonization policies. In comparison, the upper bound above 690 TWh suggests high economic and population growth, accompanied by slower technology adoption.

By explicitly quantifying forecast uncertainty, this analysis provides critical input for scenario planning and infrastructure investment decisions. Energy planners can utilize this probabilistic range to design flexible policies, stress-test grid capacity, and allocate resources across high-, medium-, and low-growth futures.

While this Monte Carlo framework improves upon deterministic forecasts, it is limited by the assumption of normally distributed inputs and a fixed interaction structure. Future research could incorporate more advanced techniques, such as Latin Hypercube Sampling or Bayesian methods, and extend the model to include correlated variables and non-linear feedback mechanisms.

4.2. Scenario-Based Sensitivity Analysis

The GDP growth rates selected for the scenario-based sensitivity analysis—2.0%, 3.5%, and 5.0% annually—are grounded in both historical economic data and national policy documents. The baseline scenario (3.5%) reflects Turkey’s average annual GDP growth between 2002 and 2022, which ranged between 3.3% and 5.0% depending on the sub-period and macroeconomic context [9,13]. The low-growth scenario (2.0%) simulates

a stagnation environment similar to periods of global economic downturn, pandemic recovery, or domestic fiscal tightening. In contrast, the high-growth scenario (5.0%) aligns with the macroeconomic assumptions articulated in the Eleventh Development Plan of Turkey, which targets an average growth rate of 5.0% in the medium term.

This scenario structure is also consistent with international energy modeling practices, including those adopted by the International Energy Agency and the European Commission's long-term energy outlooks, where low, base, and high economic growth trajectories are typically used to test the robustness of energy forecasts under uncertainty.

To address the forecast's sensitivity to macroeconomic and policy uncertainty, a fundamental scenario-based sensitivity analysis was conducted. Three GDP growth scenarios were defined:

- Low-growth scenario (2.0% annual GDP increase)—representing economic slowdown or policy stagnation;
- Baseline scenario (3.5%)—based on historical averages and current policy trends;
- High-growth scenario (5.0%)—reflecting rapid development and strong electrification efforts.

Electricity demand was projected to 2045 under each scenario using the multiple linear regression model. The corresponding demand values were as follows:

- Low growth: 712 TWh;
- Baseline: 778 TWh;
- High growth: 851 TWh.

These results suggest that a $\pm 1.5\%$ shift in the average GDP growth rate could lead to a $\sim 9\%$ swing in projected demand by 2045. This highlights the model's sensitivity to economic assumptions and underscores the importance of integrating uncertainty bands or scenario envelopes into long-term energy planning.

Future studies can incorporate additional scenario dimensions, such as population dynamics, technological adoption rates, and policy incentives, to create a more comprehensive and robust framework.

5. Conclusions

This study examined the long-term electricity demand in Turkey using various forecasting methods, including linear regression, polynomial regression, multiple linear regression, and artificial neural networks (ANN). The results indicated that electricity demand is mainly influenced by industrial activity, population growth, and economic development. While the statistical models closely matched historical trends, the artificial neural network provided more conservative forecasts and showed lower error metrics, such as root mean square error (RMSE), highlighting its reliability in unstable conditions.

Based on these findings, the following policy recommendations propose to support Turkey's sustainable energy transition:

- Invest in renewable energy infrastructure to reduce dependency on fossil fuels and ensure long-term supply security.
- Support the deployment of electric vehicles (EVs) by expanding incentive programs and nationwide charging networks.
- Enhance energy efficiency programs across residential, commercial, and industrial sectors.
- Promote digitalization and intelligent energy systems, including real-time monitoring and demand-side management tools, to enhance energy efficiency.
- Diversify energy sources by leveraging domestic coal, geothermal, and hydroelectric potential alongside renewable energy sources.

- Develop adaptive forecasting systems that integrate real-time data and account for economic, technological, and policy-driven shifts.
- Increase public awareness through energy literacy campaigns and behavior-oriented sustainability incentives.

These measures are crucial for developing a resilient, efficient, and low-carbon energy system that can meet Turkey's growing demand while upholding its environmental commitments. By providing a comparative assessment of forecasting methodologies and identifying the key factors driving demand, this study offers valuable insights into Turkey's efforts to achieve energy security, economic resilience, and environmental sustainability.

Although this study presents important findings regarding Turkey's long-term electricity demand using various forecasting approaches, a notable limitation is the lack of direct alignment between the model-based projections and the recommended policy actions. Suggestions such as promoting renewable energy, expanding electric vehicle (EV) infrastructure, and enhancing energy efficiency remain broad and are not explicitly linked to the forecast trajectories outlined in the study.

The substantial divergence between the ANN-based forecast (516.36 Mtoe) and regression-based projections (approximately 606.00 Mtoe) holds important implications for national energy policy and infrastructure planning. If the lower-end forecast materializes, it would indicate the successful implementation of energy efficiency measures, behavioral demand management, and structural shifts, such as the adoption of distributed renewables and the electrification of end-use sectors. Under this scenario, investment priorities would center on innovative grid technologies, demand response systems, and decentralized generation. Conversely, the higher-end projection would necessitate significant expansion in generation capacity, bulk transmission infrastructure, and fossil fuel imports, assuming continued reliance on traditional consumption patterns. This scenario would require aggressive capital investment in utility-scale renewables, new thermal plants, and grid reinforcement.

The contrast highlights the need for scenario-linked policy frameworks that can adapt to either pathway. Therefore, integrating energy forecasting with policy contingency planning is critical for maintaining supply security while pursuing decarbonization objectives. Aligning each forecast scenario with actionable investment and regulatory strategies can help decision-makers better prepare for demand-side volatility and systemic shifts in Turkey's energy landscape.

The projected demand range for 2045, spanning from 516.36 Mtoe (ANN model) to 606.00 Mtoe (regression-based models), reflects a substantial divergence in possible future consumption levels. However, the manuscript does not elaborate on how these scenarios correspond to distinct policy or investment pathways. For instance, the lower-end forecast of 516.36 Mtoe could represent an outcome achievable under the successful implementation of the National Energy Efficiency Action Plan and the Energy Efficiency Strategy Document, which target measurable reductions in consumption across residential, industrial, and transport sectors. Conversely, the higher-end projections imply significant needs for expanding generation capacity, accelerating grid modernization, and addressing peak demand through flexible and resilient infrastructure.

The range of forecasts generated by different models has significant implications for Turkey's energy infrastructure planning and policy orientation. Under lower-demand scenarios (e.g., ANN: ~615 TWh by 2045), the focus may shift toward optimizing existing capacity, enhancing grid efficiency, and investing in demand-side management. Conversely, higher-demand projections (e.g., MLR: ~850 TWh) imply the need for accelerated expansion of generation capacity, particularly in renewables, modernization of transmission

infrastructure, and scaled-up electrification initiatives, including electric vehicles (EVs) and green hydrogen.

Policy design under such uncertainty must incorporate flexibility, adaptive targets, and robust scenario planning mechanisms to accommodate a broad spectrum of potential outcomes. Over-investment in high-demand assumptions or under-preparation for potential demand surges both carry significant risks, including stranded assets or supply shortfalls.

The scenario analysis reveals significantly different planning implications across demand trajectories. In the low-demand scenario (e.g., under 700 TWh by 2045), policy focus may shift toward maximizing existing capacity utilization, enhancing grid efficiency, and delaying large-scale infrastructure investments. Conversely, in the high-demand scenario (above 850 TWh), the accelerated deployment of renewable generation, grid expansion, and scaling of EV infrastructure would be critical. These contrasting responses underscore the need for adaptive, scenario-informed policy frameworks that incorporate demand uncertainty into investment, generation planning, and decarbonization pathways.

In addition to these deterministic estimates, a Monte Carlo simulation was implemented to assess forecast uncertainty. This simulation introduced random variation into key macroeconomic and structural variables, yielding a broader range of projected outcomes. The results suggest that electricity demand in 2045 could fall between 552 TWh and 1193 TWh with a 90% confidence interval and a mean value of 840 TWh. These findings underscore the need for flexible and resilient energy planning strategies that can accommodate a wide range of potential future demand scenarios.

Moreover, the absence of scenario-based sensitivity analysis—such as variations in GDP growth, renewable energy penetration, population trends, or electrification of end-use sectors—limits the interpretability and policy relevance of the findings. Previous research emphasizes that model outputs should be tested against critical assumptions to enhance their robustness and decision-making utility [23,30]. Similarly, effective policy design requires empirical grounding in model-specific trajectories rather than normative generalizations [28].

To strengthen the link between empirical forecasts and strategic planning, future iterations of this research should consider the following:

- Establishing explicit correspondences between each demand scenario and the infrastructural or regulatory actions required to support it.
- Quantifying the effects of targeted policies (e.g., widespread EV adoption and smart grid deployment) on reducing or reshaping demand.
- Conducting systematic sensitivity analyses to assess the responsiveness of forecasts to macroeconomic, demographic, and technological variables.

Integrating cost-effectiveness metrics to prioritize interventions based on their marginal impact on demand stabilization and carbon reduction.

Such enhancements would improve the manuscript's utility not only as an academic exercise but also as a practical decision-support tool for policymakers engaged in Turkey's energy transition. By aligning model-based evidence with structured policy pathways, the study can provide more informed guidance for sustainable, resilient, and data-driven national energy strategies.

Author Contributions: Methodology, E.K.; Investigation, S.K.K.; Data curation, S.K.K.; Writing—original draft, S.K.K.; Writing—review & editing, E.K.; Supervision, E.K.; Funding acquisition, E.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Turkish Ministry of Energy and Natural Resources. Türkiye Ulusal Enerji Planı. 2022. Available online: <https://enerji.gov.tr/duyuru-detay?id=20317> (accessed on 19 January 2023).
2. Whiteman, A.; Akande, D.; Elhassan, N.; Escamilla, G.; Ahmed, I. International Renewable Energy Agency, Abu Dabi. Renewable Energy Statistics 2023. IRENA. Available online: <https://www.ourenergypolicy.org/resources/renewable-energy-statistics-2023/> (accessed on 29 July 2023).
3. Chamber of Mechanical Engineers. Energy Outlook Report. 2021. Available online: <https://mmo.org.tr/merkez/haber/turkiyenin-enerji-gorunumu-nisan-2021-sunumu-yayinlandi> (accessed on 8 April 2021).
4. Turkish Ministry of Energy and Natural Resources. Natural Gas Information Center. 2023. Available online: <https://enerji.gov.tr/bilgi-merkezi-enerji-dogalgaz> (accessed on 19 June 2023).
5. Turkish Ministry of Energy and Natural Resources. Oil Information Center. 2023. Available online: <https://enerji.gov.tr/bilgi-merkezi-enerji-petrol> (accessed on 19 June 2023).
6. Turkish Ministry of Energy and Natural Resources. *Turkish Hard Coal Enterprises Activity Report*; Turkish Ministry of Energy and Natural Resources: Ankara, Turkey, 2023.
7. Turkish Ministry of Energy and Natural Resources. Electricity Information Center. Available online: <https://enerji.gov.tr/bilgi-merkezi-enerji-elektrik> (accessed on 25 November 2024).
8. Turkish Ministry of Energy and Natural Resources. *Turkey Electricity Demand Projection Report Covering the Years 2019–2039*; Turkish Ministry of Energy and Natural Resources: Ankara, Turkey, 2019.
9. Turkish Statistical Institute. Population Projection, 2023–2100. 2023. Available online: <https://data.tuik.gov.tr/Bulten/Index?p=Population-Projections-2023-2100-53699&dil=2> (accessed on 30 July 2024).
10. Turkey Electricity Transmission Corporation. *Turkey Electricity Production-Transmission. 2020 Statistics*; Turkey Electricity Transmission Corporation: Ankara, Turkey, 2023.
11. Turkish Statistical Institute. Motor Vehicle. 2024. Available online: <https://data.tuik.gov.tr/Bulten/Index?p=Motorlu-Kara-Tasitlari-Kasim-2024-53462> (accessed on 20 December 2024).
12. Turkish Ministry of Energy and Natural Resources. Energy Efficiency 2030 Strategy and National Energy Efficiency Action Plan 2024–2030. 2022. Available online: <https://enerji.gov.tr/Media/Dizin/EVCED/tr/EnerjiVerimlili%C4%9Fi/UlusalEnerjiVerimlili%C4%9FiEylemPlan%C4%B1/Belgeler/EnerEffi2030Str2ndNatEnerEffiActPlan2024-2030.pdf> (accessed on 19 January 2023).
13. International Monetary Fund. World Economic Outlook October 2024. Available online: <https://www.imf.org/en/Publications/WEO/Issues/2024/10/22/world-economic-outlook-october-2024> (accessed on 19 January 2023).
14. Lewis, C.D. Short-term forecasting for stationary demand situations. In *Demand Forecasting and Inventory Control*; Routledge: London, UK, 1998. [CrossRef]
15. Turkish Statistical Institute. Annual Industry and Statistics 2019. 2019; p. 71. Available online: <https://data.tuik.gov.tr/bulten/Index?p=Yillik-sanayi-veHizmet-Istatistikleri-2019-33761> (accessed on 19 June 2023).
16. Ugbebe, P.O.; Diemuodeke, O.E.; Aikhuele, D.O. Electricity Demand Forecasting Methodologies and Applications: A Review. *Sustain. Energy Res.* **2025**, *1*, 100149. [CrossRef]
17. Abisoye, O.; Sun, Y.; Zenghui, W. A survey of artificial intelligence methods for renewable energy forecasting: Methodologies and insights. *Renew. Energy Focus* **2023**, *48*, 100529. [CrossRef]
18. Gültekin, Ö. Bursa İli Orta Dönem Elektrik Talep Tahmini. Master’s Thesis, Dumlupınar University, Kütahya, Turkey, 2009.
19. Karaca, A. 2000 Sonrası Türkiye’nin Enerji Bağımlılığı ve Dış Ticaretin Ekonomik Analizi. Master’s Thesis, Marmara University, Institute of Social Sciences, Istanbul, Turkey, 2019.
20. Koç, E.; Şenel, M.C. Dünyada ve Türkiye’de Enerji Durumu. *Mühendis ve Makina* **2015**, *54*, 32–44.
21. Republic of Turkey, Ministry of Environment and Urbanization. Climate Change and Sustainable Development Policies. 2021. Available online: <https://www.csb.gov.tr> (accessed on 13 May 2025).
22. Akkuyu Nuclear JSC. Akkuyu Nuclear Power Plant—A Clean Energy Solution for Turkey. 2024. Available online: <https://www.akkunpp.com> (accessed on 13 May 2025).
23. Suganthi, L.; Anand, A.S. Energy models for demand forecasting—A review. *Renew. Sustain. Energy Rev.* **2012**, *16*, 1223–1240. [CrossRef]
24. Wang, J.; Li, Y.; Mai, J.; Yuan, M.; Liu, Z. Highway Typical Scenario Operation and Maintenance Energy Demand Forecasting. *Sustainability* **2025**, *17*, 1929. [CrossRef]
25. Vakhnin, A.; Ryzhikov, I.; Niska, H.; Kolehmainen, M. A Novel Multi-Objective Hybrid Evolutionary-Based Approach for Tuning Machine Learning Models in Short-Term Power Consumption Forecasting. *AI* **2024**, *5*, 2461–2496. [CrossRef]
26. Hong, T.; Pinson, P.; Wang, Y.; Weron, R.; Yang, D.; Zareipour, H. Energy Forecasting: A Review and Outlook. *IEEE Open Access J. Power Energy* **2020**, *7*, 376–388. [CrossRef]

27. Divina, F.; Torres, M.G.; Vela, F.A.G.; Noguera, J.L.V. A Comparative Study of Time Series Forecasting Methods for Short Term Electric Energy Consumption Prediction in Smart Buildings. *Energies* **2019**, *12*, 1934. [[CrossRef](#)]
28. Carley, S. State Renewable Energy Electricity Policies: An Empirical Evaluation of Effectiveness. *Energy Policy* **2009**, *37*, 3071–3081. [[CrossRef](#)]
29. Daut, M.A.M.; Hassan, M.Y.; Abdullah, H.; Rahman, H.A.; Abdullah, M.P.; Hussin, F. Building Electrical Energy Consumption Forecasting Analysis Using Conventional and Artificial Intelligence Methods: A Review. *Renew. Sustain. Energy Rev.* **2017**, *70*, 1108–1118. [[CrossRef](#)]
30. Debnath, K.B.; Mourshed, M. Forecasting Methods in Energy Planning Models. *Renew. Sustain. Energy Rev.* **2018**, *88*, 297–325. [[CrossRef](#)]
31. Arslan, E.; Solak, A. Türkiye’de Yenilenebilir Enerji Tüketiminin İthalat Üzerindeki Etkisi. *Uluslararası Toplum Araştırmaları Derg.* **2019**, *10*, 1380–1407. [[CrossRef](#)]
32. Bayraç, H.N. Küresel Enerji Politikaları ve Türkiye: Petrol ve Doğal Gaz Kaynakları Açısından Bir Karşılaştırma. *Eskişehir Osman. Üniv. Sos. Bilim. Derg.* **2009**, *10*, 115–142.
33. Phuangpornpitak, N.; Prommee, W. A Study of Load Demand Forecasting Models in Electric Power System Operation and Planning. *GMSARN Int. J.* **2016**, *10*, 19–24.
34. Aderibigbe, A.O.; Ani, E.C.; Ohenhen, P.E.; Ohalet, N.C.; Daraojimba, D.O. Enhancing Energy Efficiency with AI: A Review of Machine Learning Models in Electricity Demand Forecasting. *Energy Sustain. Technol. J.* **2023**, *4*, 341–356.
35. Turkish Statistical Institute. Statistics on Final Energy Consumption in Industry, 2023. 2024. Available online: <https://data.tuik.gov.tr/Bulten/Index?p=Statistics-on-Final-Energy-Consumption-in-Industry-202353806&dil=2> (accessed on 1 November 2024).
36. Akpınar, A.; Kömürcü, M.İ.; Kankal, M.; Özölçer, İ.H.; Kaygusuz, K. Energy Situation and Renewables in Turkey and Environmental Effects of Energy Use. *Renew. Sustain. Energy Rev.* **2008**, *12*, 2013–2039. [[CrossRef](#)]
37. Arnob, S.S.; Arefin, A.I.M.S.; Saber, A.Y.; Mamun, K.A. Energy Demand Forecasting and Optimizing Electric Systems for Developing Countries. *IEEE Access* **2023**, *11*, 39751–39775. [[CrossRef](#)]
38. Zhussupbekov, M.; Memon, S.A.; Khawaja, S.A.; Nazir, K.; Kim, J. Forecasting Energy Demand of PCM Integrated Residential Buildings: A Machine Learning Approach. *Energy Rep.* **2023**, *9*, 1234–1245. [[CrossRef](#)]
39. Guo, N.; Chen, W.; Wang, M.; Tian, Z.; Jin, H. Applying an Improved Method Based on ARIMA Model to Predict the Short-Term Electricity Consumption Transmitted by the Internet of Things (IoT). *Wirel. Commun. Mob. Comput.* **2021**, *2021*, 6610273. [[CrossRef](#)]
40. Mir, A.A.; Alghassab, M.; Ullah, K.; Khan, Z.A.; Lu, Y.; Imran, M. A Review of Electricity Demand Forecasting in Low- and Middle-Income Countries: The Demand Determinants and Horizons. *Sustainability* **2020**, *12*, 5931. [[CrossRef](#)]
41. Pérez-García, J.; Moral-Carcedo, J. Analysis and Long-Term Forecasting of Electricity Demand Through a Decomposition Model: A Case Study for Spain. *Energy* **2016**, *97*, 127–143. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.

Copyright of Energies (19961073) is the property of MDPI and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.